Genetic Endowments and the Long-Run Health Returns to Compulsory Schooling

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November 10, 2026

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Abstract

Genetic differences tied to educational attainment predict later-life success and wellbeing. I study whether they also moderate the causal effect of education on physical health in old age. Using data on individuals aged 65 to 85 from the English Longitudinal Study of Ageing, I exploit the 1947 raising of the school leaving age in the UK from 14 to 15 in a fuzzy regression discontinuity setting to estimate the causal effect of schooling on frailty, the share of age-related health deficits. I estimate interactions with an educational attainment polygenic index (EA PGI), a DNA-based measure of genetic propensity for education. Health returns to schooling are concentrated at the top of the EA PGI distribution. In the highest quintile, additional education reduces frailty by 6 percentage points, compared with near-zero effects at the bottom and possibly small benefits in the middle. This suggests a complementarity between genetic endowments and education. These health improvements originate at the upper part of the frailty distribution. I further show that the EA PGI also affects educational responses to the schooling reform. Using marginal treatment effects, I find that the main results are not driven by these differences in compliance behavior or unobserved characteristics of compliers.

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I gratefully acknowledge financial support from the German Research Foundation (DFG, project number 437564156)

1 Introduction

Genetic differences between individuals are important drivers of success and wellbeing (Benjamin et al. 2012; Biroli et al. 2026). Their influence is already apparent in school, where achievement gaps by genetic propensity for education are as large as socioeconomic gaps (Houmark et al. 2025). Children with more of these genetic markers develop stronger cognitive and noncognitive skills (Belsky et al. 2016; Houmark, Ronda, and Rosholm 2024). Their influence also extends into adulthood and old age. Genetic propensity for education is correlated with occupation, income, and wealth (Papageorge and Thom 2020; Carvalho 2025), personality (Mõttus et al. 2017), health behaviors such as smoking (Wedow et al. 2018), and several severe diseases (Okbay et al. 2022).

Genetic propensity for education also affects later-life outcomes by moderating the effects of environments tied to education and skill accumulation. Children with at the higher end of the distribution acquire and retain skills more effectively, so that early advantages accumulate (Houmark, Ronda, and Rosholm 2024). They likely face lower marginal costs of schooling (Carvalho 2025) and therefore obtain more education (Hollenbach, Schmitz, and Westphal 2026). Furthermore, parents invest more in children with higher genetic propensity for education, reinforcing initial differences (Houmark, Ronda, and Rosholm 2024). Additionally, it affects the returns to different kinds of investments. For example, an additional year of schooling raises income and old-age cognitive abilities mainly for individuals with high propensity for education (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026).

In this paper, I study how the effect of education on physical health in old age varies with genetic propensity for educational attainment. Using data on individuals aged 65–85 from the English Longitudinal Study of Ageing, I exploit the 1947 raising of the UK school leaving age from 14 to 15 in a fuzzy regression discontinuity design to estimate the causal effect of schooling. I measure health with a frailty index, the share of age-related health deficits present in an individual. To quantify genetic propensity, I use an educational attainment polygenic index (EA PGI), a DNA-based measure that sums the genetic variants an individual carries, each weighted by how strongly it is associated with schooling. It predicts a sizable share of the variation in educational attainment between individuals, but is not deterministic.

This paper contributes to the literature in two ways. First, I investigate the role of the EA PGI in health at old-age. I am the first to study the effect of interactions of educational attainment and genetic propensity for educational attainment in old-age health. Related studies on differential effects of education on later-life outcomes by genetic predisposition often find that genes and education reinforce each other. For health, the gains from education are largest among individuals at high genetic *risk* for the specific condition, including Alzheimer's disease and mid-life overweight (Barcellos et al. 2025; Barcellos, Carvalho, and Turley 2018), while the effects on metabolic health in middle age do not differ by the EA PGI (Barcellos, Carvalho, and Turley 2018). In contrast, the effects of education on income and on cognitive ability in old age increase in the EA PGI (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026). I add to this by showing that an early-life interaction

of an exogenous increase in schooling and genetic propensity contributes to differences in health late in life.

Second, I contribute to the question of whether the typically large descriptive health differences by education reflect a causal effect, and if so, for whom and in which contexts. Evidence from UK compulsory schooling reforms is mixed. Clark and Royer (2013) estimate very small effects of the 1947 and 1972 reforms on self-rated health, and Jürges, Kruk, and Reinhold (2013) find no effects on self-rated health or biomarkers for inflammation. For the 1972 reform, Gehrsitz and Williams (2024) report no effect on subjective health but fewer hospitalizations among men, and Janke et al. (2020) document a lower probability of diabetes. Silles (2009) shows that education significantly increases self-reports of good health and of being free of (limiting) illness. Studies using UK Biobank data on the 1972 reform find that the additional education reduced body size at the upper end and increased blood pressure at the lower end of the respective distributions (Barcellos, Carvalho, and Turley 2023), and lowered the risks of diabetes and mortality (Davies et al. 2018). Except for Jürges, Kruk, and Reinhold (2013), these studies use cohorts that were still relatively young at the time, and all of them examine specific health measures, often chronic diseases or self-rated health. I instead study individuals up to age 85 and use the frailty index, which combines up to 56 diagnoses, conditions, limitations, and tests into a comprehensive measure of mostly functional health in old age.

My results first establish a sizable genetic health gap in old age. The difference in frailty between the highest and lowest quintile of the EA PGI is about as large as the difference between the first and the fourth quintile of socioeconomic status measured using an index including educational attainment, pension income and household wealth. Furthermore, frailty is lower among the more educated in every quintile of the EA PGI, but the health advantage of more schooling is concentrated with those in the highest PGI quintile. My fuzzy regression discontinuity estimates suggest that this pattern is also reflected in the causal effect of education. I find that health returns to an additional year of schooling rise monotonically along the EA PGI, from near zero in the lowest quintile to a 6 percentage point reduction in frailty in the highest. This gradient points to a complementarity between genetic endowments and education in old-age health. The gains come from the upper part of the frailty distribution. Additional schooling lowers the probability of being frail and I find suggestive evidence that for those in the highest PGI quintile, it also increases the probability of remaining nearly deficit-free for longer.

I further show strong differences in educational choices in response to the schooling reform. The share of always-takers to the reform increases along the EA PGI, while the share of compliers decreases. This suggests that genetic endowments associated with educational attainment contribute to such educational choices. However, these differences may also drive my interaction effects, because they indicate that compliers in different parts of the EA PGI distribution may not be comparable. Estimating marginal treatment effects (MTEs), the treatment effects at a fixed level of unobserved resistance to schooling, fixes this by explicitly modeling educational choice as a cost-benefit tradeoff that depends on the EA PGI. Following Mogstad, Santos, and Torgovitsky (2018), I estimate bounds on the MTEs by finding optimal flexible polynomials that are consistent with the data. In this way, I cleanly separate the

interaction effects of schooling and the EA PGI from treatment effect heterogeneity along unobserved resistance to schooling. I find suggestive evidence that differences in complier characteristics may have led to a lower 2SLS estimate in the lowest EA PGI quintile. The gradient of the effect of education on frailty along the EA PGI may be less steep than 2SLS implies. Nevertheless, the MTE estimates confirm that by far the largest health returns to education are concentrated among those in the highest EA PGI quintile, which means that my main result is not just driven by differences in unobserved complier characteristics that also determine compliance behavior.

Investigating potential channels of an average effect of education on frailty, I show that the additional education from the reform improved smoking behavior and physical activity in old age and reduced the probability of receiving a disability pension. I do not find any impacts on old-age pensions, income or employment history.

The 1947 compulsory schooling reform increased educational attainment most among students with a low EA PGI. Since such reforms are targeted at those who leave school early, the reform seems to have largely succeeded. However, the additional education improved later-life health only for those with higher PGI scores. This suggests that the reform contributed to a widening of later-life health differences. While compulsory schooling seems to be able to compensate for certain genetic risk factors, like those related to overweight (Barcellos, Carvalho, and Turley 2018), it exacerbates existing differences in the EA PGI, which has been shown to be a relevant source of inequality in school (Houmark et al. 2025; Biroli et al. 2026). Nevertheless, this does not mean that policy changes at other educational margins would not be able to improve later-life health of low-PGI students. One such margin could be school quality, which the 1947 reform did not improve (Clark 2023). Improvements in school quality could benefit later-life outcomes of low-PGI students, since higher-quality schools have been shown to improve academic achievement most for those at the lower end (Arold, Hufe, and Stoeckli 2025).

The remainder of this paper is structured as follows. Section 2 introduces the institutional setting of the 1947 UK compulsory schooling reform and the measure of genetic propensity for educational attainment. Section 3 describes the data and sample, the variables, and the empirical methods and assumptions. Section 4 presents the main results. I first show descriptive patterns, then estimate the interaction causally, and estimate marginal treatment effects. I close the section with robustness checks and suggestive evidence on non-health channels. Section 4.8 concludes.

2 Institutional setting and background

2.1 The 1947 raising of the school leaving age

To identify the causal effect of schooling on late-life health, I exploit the 1947 compulsory schooling reform in the UK. A second reform raised the leaving age from 15 to 16 in September 1972. I do not use this reform because cohorts affected by it are still too young in the data to study late-life health outcomes. Furthermore, genetic data have only been collected once in the English Longitudinal Study of Ageing, covering mostly the older cohorts.

The 1944 Education Act raised the minimum school-leaving age from 14 to 15, with implementation set by Ministerial order to April 1, 1947 (Clark and Royer 2013). Because children in the UK typically started school at age five, the reform extended the mandatory length of schooling from nine to ten years. Cohorts born from April 1933 onward, the pivotal cohort, were the first bound by the new rule. I later on exploit this discontinuity by comparing cohorts born before and after the cutoff.

The reform had a strong and broad effect on educational attainment, shifting a large share of the students from nine to ten years of schooling. Figure 1 illustrates this shift using cohort-level aggregates from ELSA. The figure plots the share of individuals still enrolled at ages 15 and 16 by birth cohort, with a vertical line marking the pivotal cohort. The top line (circles) tracks enrollment at age 15. For unaffected cohorts, this share sits around 40%, implying that roughly 60% of students left at the previous compulsory leaving age of 14. Both series trend upward in the cohorts just before the cutoff, but only enrollment at 15 jumps sharply at the pivotal cohort, rising from around 60% to nearly 100%. Enrollment at age 16 (diamonds) shows no such discontinuity. Compliance was high, but not complete. Some individuals report leaving school at age 14 even after the reform took effect. As Clark and Royer (2013) documents, this noncompliance is concentrated among individuals born in the summer months, who turned 15 between the end of one school year and the start of the next, and could therefore leave at age 14.

The reform raised the quantity of schooling without changing its quality. Nearly all compliers attended lower-track schools, which emphasized practical education and were of lower quality than grammar schools in terms of class size and teacher qualifications (Clark 2023). School resources adjusted to accommodate the increased enrollment, so these inputs did not change as a result of the reform (Clark and Royer 2013). The additional year did not raise the probability of obtaining formal academic or vocational qualifications on average (Clark 2023). Instead, schools used the extra year to extend the existing curriculum, introducing some students to more advanced material and helping others consolidate basic skills (Clark and Royer 2013). The effect of the reform can therefore be thought of as an increase in quantity of schooling.

The 1947 reform has been used to identify causal effects of education on wages (Harmon and Walker 1995; Oreopoulos 2006; Devereux and Hart 2010; Clark and Royer 2013), other labor market outcomes (Clark 2023), financial behavior (Gray, Montagnoli, and Moro 2021), health (Silles 2009; Powdthavee 2010; Clark and Royer 2013; Jürges, Kruk, and Reinhold 2013), intergenerational health (Lindeboom, Llena-Nozal, and van der Klaauw 2009; Silles 2011), health knowledge (Johnston et al. 2015), mortality (Clark and Royer 2013; Gathmann, Jürges, and Reinhold 2015), and cognitive ability (Banks and Mazzonna 2012; Hollenbach, Schmitz, and Westphal 2026).

2.2 Measuring genetic propensity for education

Frequently used measures of student ability use final products of that ability, such as IQ tests results, scores on academic or vocational tests, and college admission decisions (see, e.g., Brinch and Galloway 2011; Neal and Johnson 1996; Black and Smith 2006; Dale and Krueger 2002). Perhaps more cautiously interpreted as a propensity for education or learning, these measures also have in common that they

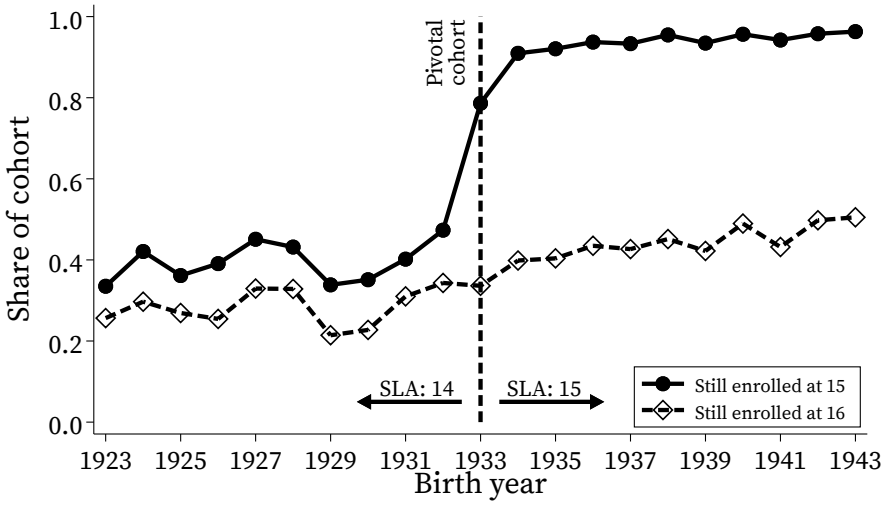


FIGURE 1 – School enrollment at ages 15 and 16 by birth cohort

Note: The 1944 Education Act raised the minimum school leaving age (SLA) from 14 to 15 for cohorts born from April 1933 onward. This figure shows yearly cohort means for enrollment at 15 and 16 since information on month of birth are unavailable. The mean of the pivotal cohort of 1933 therefore includes both individuals affected and unaffected by the schooling reform. The share still enrolled at age 16 is shown for comparison. Sample: 10,709 individuals across waves 1-10 of the English Longitudinal Study of Ageing.

are malleable outputs of complex inputs early in life, such as genes, conditions and investments (Todd and Wolpin 2003; Cunha and Heckman 2007).

In this paper, I rely on a DNA-based measure that has the advantage of being fixed at conception and not modifiable afterwards. Its use for complex traits like educational attainment is grounded in two basic facts about human genetics. First, humans share most of their genome, but small differences can explain variation in traits. The human genome consists of roughly three billion DNA base pairs, and any two individuals share more than 99 percent of their DNA on average (Auton et al. 2015; Nurk et al. 2022). The remaining differences account for a substantial share of the variation in observable traits and behavior: a meta-analysis of twin studies by Polderman et al. (2015) finds that genetic factors explain about half of the population variation across 2,929 traits. Second, There is no single gene for a complex trait. Genetic influence on traits and behavior is mostly polygenic, meaning it is spread across many locations in the genome that each contribute a small effect (Chabris et al. 2015; Visscher et al. 2017). Education fits this pattern, with thousands of genetic variants individually associated with years of schooling, each contributing only a small share (Lee et al. 2018; Okbay et al. 2022).

The standard tool for summarizing small additive contributions of many genetic variants into one measure is the polygenic index (PGI). Most genetic differences between individuals come down to single positions in the genome where the DNA base varies across the population. These are called single-nucleotide polymorphisms (SNPs). A PGI summarizes an individual's genetic predisposition for a given outcome by summing their SNPs, each weighted by its estimated association with that outcome. To quantify SNPs, one variant present in the population is chosen as the reference variant. Any person i can inherit up to two copies of each SNP j, one from each parent and therefore carries either zero,

one, or two copies of the reference variant $r_{ij} \in \{0, 1, 2\}$. Weights w_j, i.e. the association of each SNP with the outcome, are obtained in large discovery studies, called genome-wide association studies (GWAS). They estimate how strongly each SNP is correlated with the trait of interest by regressing the trait on each SNP, one at a time, in a large separate sample to avoid overfitting.

The PGI is then simply the weighted sum of counts of the reference variant across all SNPs:

$$PGI_i = \sum_j w_j r_{ij} \tag{1}$$

The PGI is a single individual-level index of additive genetic probability for a certain outcome. In this paper, I use a PGI for educational attainment (EA PGI). It is approximately normally distributed, with individuals at the upper end carrying more of the alleles that, on a population level, are positively associated with schooling and those at the lower end carrying fewer. Even though, at the individual level, it is informative about educational attainment, the index is not deterministic, because its realization depends on early-life environments. Individuals with a high EA PGI can have low attainment, and vice versa. Therefore, the EA PGI can not directly be interpreted as innate ability. Furthermore, it captures more than ability, including traits like motivation or part of family environments (Kong et al. 2018; Demange et al. 2021). Instead, it can be understood as an index of genetic factors relevant for human capital accumulation (Barth, Papageorge, and Thom 2020).

3 Data and methods

3.1 Data and variables

I use data from the the English Longitudinal Study of Ageing (ELSA), a nationally representative longitudinal survey of adults aged 50 and older living in England (Banks et al. 2026). ELSA has been conducted biennially since 2002 and includes detailed information on physical and mental health, socioeconomic status, and, over time, has integrated biological and genetic data. For more information, see Steptoe et al. (2013).

Health in old age. My outcome variable is the frailty index, the share of potential age-related health deficits that an individual exhibits in a given survey wave. Deficits are symptoms, signs, diagnosed diseases, disabilities, and functional impairments. The index therefore summarizes health across many dimensions in a single number bounded between zero and one. It is grounded in the view of aging as an accumulation of deficits and a gradual loss of the body's redundancies (Mitnitski, Mogilner, and Rockwood 2001; Dalgaard and Strulik 2014). A higher value indicates a more advanced state of aging rather than the presence of one particular condition. A key characteristic of the frailty index is its ability to capture health status and its variation with high granularity. Compared with binary or ordinal health measures common in applied work, it is continuous and finely graded, and it relies on many concrete items instead of a single overall self-assessment (Hosseini, Kopecky, and Zhao 2022).

The frailty index is an established measure of health. First introduced in gerontology (Mitnitski, Mogilner, and Rockwood 2001) and standardized by Searle et al. (2008), it predicts mortality, institutionalization, and further adverse health events, and it predicts these outcomes more accurately than self-reported health status (Rockwood and Mitnitski 2007; Hosseini, Kopecky, and Zhao 2022). Results also do not depend on the exact items chosen. As long as enough deficits from different areas of health enter, indices built from different item lists behave similarly (Searle et al. 2008; Hosseini, Kopecky, and Zhao 2022). Economists have taken up the measure both as the health state in models of aging (Dalgaard and Strulik 2014; Strulik 2018b) and as the health outcome in empirical work, in studies of aging patterns across Europe (Abeliansky and Strulik 2018), of retirement and occupational health gaps (Abeliansky and Strulik 2023), and of the role of health in life-cycle earnings inequality (Hosseini, Kopecky, and Zhao 2022, 2025).

I follow the standard procedure of Searle et al. (2008), implemented in ELSA, for example by Marshall et al. (2015) and Niederstrasser, Rogers, and Bandelow (2019), and use a broad set of 56 deficits available in ELSA. They include diagnoses of serious illnesses, difficulties with activities of daily living and mobility, sensory impairment, self-rated health, depressive symptoms, and results from cognitive tests. Each item takes a value between zero and one. Binary items, such as diagnoses, equal one if the deficit is present and zero otherwise. Ordinal items, such as self-rated health, can also take intermediate values. The index is the sum of all 56 items divided by 56. For instance, an individual with 14 of the 56 deficits has a frailty index of 0.25. Appendix Table A.1 lists all items and how each is coded.

Education. ELSA does not record years of schooling directly. Instead, it reports the age at which a respondent left full-time education. My measure of educational attainment is an indicator variable equal to 0 if an they left school at age 14 or below and 1 if they stayed until at least age 15, i.e., left school at 15 or older. This is the margin targeted by the compulsory schooling reform. Most individuals in the final sample have low educational attainment, which was typical for these birth cohorts and one reason for the compulsory schooling reform (Oreopoulos 2006; Woodin, McCulloch, and Cowan 2013). Twenty-one percent left school at age 14 or earlier (the compulsory schooling age before the reform), 35 percent at age 15 (the compulsory schooling age after the reform), 30 percent at ages 16 to 18, and 13 percent at age 19 or later.

Schooling reform. The compulsory schooling reform took effect in April 1947 and applied to everyone born on or after 1 April 1933. Accordingly, I define the instrument that exogenously shifts educational attainment as an indicator variable equal to 0 if an individual was born before 1933 and 1 if they were born in 1933 or later. ELSA does not provide birth months, so for individuals born in 1933 it is unclear whether they were subject to the new school leaving age. Because almost all individuals are interviewed multiple times and in different months, I can combine birth year, interview month, and age at interview to infer for part of the 1933 cohort whether they were born between January and March (pre-reform) or between April and December (reform). I drop the remaining individuals born in 1933, whom I cannot definitively assign to either side of the reform.

Genetic propensity for educational attainment. To measure the genetic propensity for educational attainment, I use an EA PGI, as introduced in Section 2.2. The PGI is readily available in ELSA (NatCen Social Research 2026) and the weights used to calculate it are based on the discovery study by Lee et al. (2018) covering roughly 1.1 million individuals. Lee et al. (2018) report that it can explain 11–13 percent of the variation in educational attainment and about 10 percent of the variation in cognition in different samples.

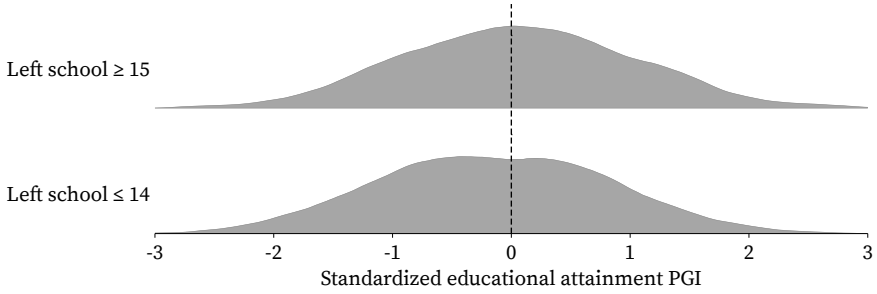


FIGURE 2 – Distribution of the EA PGI by treatment status

Note: Kernel density plots of the EA PGI distribution by educational attainment indicator (left school at 15 or later). Based on the estimation sample covering 3,415 individuals from the English Longitudinal Study of Ageing. Sample restrictions as described in Section 3.2

Figure 2 shows the distribution of the standardized EA PGI in my final sample by the educational attainment indicator I use in this paper (left school at 14 or below versus 15 or later). While both groups have individuals with high and low EA PGI and most mass in the middle, the distribution skews towards lower values in the group leaving school earlier. To investigate interactions of schooling and the PGI at different parts of its distribution, I divide the score into quintiles.

Control variables. Since I am only interested in the differential effect of education on frailty by EA PGI, it would be sufficient to use the PGI as a pre-determined (and therefore pre-treatment) interaction variable. Nevertheless, I include parental education as additional control variables, in line with Barth, Papageorge, and Thom (2020), Papageorge and Thom (2020), and Barth et al. (2022), among others.

This is because the association between an individual's EA PGI and their schooling does not reflect a clean causal effect of the PGI. Part of the association is due to genetic nurture, whereby parents pass on both genes and a home environment. Parents whose own genes are associated with more schooling also tend to provide a home environment that supports learning. The child's PGI is therefore correlated with their schooling partly because it proxies for family environment, not only because of the child's own DNA (Kong et al. 2018). Conditional on the parents' genes, the child's genes are exogenous. Therefore, controlling for parental genotype would yield a causal effect of the PGI. A second strategy is a within-family design that compares siblings. Because they share the same parents, family fixed effects absorb parental genotype. Few datasets to date contain parent-child trios or enough siblings for a well-powered analysis. In ELSA, only between-family designs are feasible. In that case, Houmark, Ronda, and Rosholm (2024) show that controlling for parental education can absorb almost all of the family environment component.

I control for parental education by including indicator variables for the age at which each parent left full-time education. In ELSA, these ages are censored at 14 and 19. This information is missing for 17.5 percent (mothers) and 18.1 percent (fathers) of my final sample. Instead of dropping these observations, I include an additional category for missing parental school-leaving age.

Furthermore, I add controls for population stratification. Because of historical migration and non-random mating, subpopulations differ systematically in the frequency of genetic variants. If they also differ in schooling or health for cultural or environmental reasons, the discovery study underlying the PGI assigns weight to variants that signal ancestry rather than the trait itself. The PGI then partly captures the environments associated with ancestry, such as region, language, or local socioeconomic conditions, in a similar way as it partly captures the family environment. I therefore follow the convention in this literature (see e.g. Barth, Papageorge, and Thom 2020; Papageorge and Thom 2020; Barth et al. 2022) and additionally control for the first ten principal components of the genetic data. These are summary indices of general genetic similarity between individuals and account for population stratification (Price et al. 2006). ELSA provides them as pre-computed variables.

3.2 Sample and summary statistics

The analysis sample includes all ELSA respondents with genetic data who were born between 1923 and 1943, i.e. within ten years of the pivotal cohort of the compulsory schooling reform. They are observed at ages 65–85 in waves 1–10 of ELSA. From the 1933 cohort itself, I drop individuals whom I cannot assign to either side of the April 1 cutoff based on birth year, interview month and reported age at interview. These restrictions leave 18,705 observations from 3,415 individuals. Section 4.6 reports robustness checks that vary these sample restrictions.

Table 1 shows summary statistics for my sample, both in its entirety and separately by treatment and control group. The average frailty index is 0.16, with a substantial standard deviation. A majority of the sample consists of treated observations and individuals born after the reform cutoff. Treatment and control group differ significantly along several dimensions. Some of these differences are mechanical, most notably birth year and being born after the cutoff. Others concern individual characteristics: the treated group has a slightly higher share of women, is slightly younger, and is more concentrated in the upper part of the EA PGI distribution. The age gap is smaller than the gap in birth years, as repeated measurements at different ages seem to offset the latter to some degree. Crucially, individuals in the treated group are significantly less frail.

3.3 Method

To estimate the causal effect of schooling on old-age frailty and how it varies with genetic propensity for education, I implement a fuzzy regression discontinuity design, estimated by two-stage least squares (2SLS). The second stage is

$$Y_{it} = \sum_{g=1}^{5} \Big(\alpha_g \, \mathbb{1}[G_i = g] + \beta_g \, E_i \times \mathbb{1}[G_i = g] \Big) + \theta_1 c_i + \theta_2 \, Z_i \times c_i + X_{it}' \gamma + \varepsilon_{it}. \tag{2}$$

TABLE 1 – Summary statistics

	Full sample		By E_i (Left school at 15 or later)		
	Mean	SD	0	1	Diff. (p-val.)
Outcome					
Frailty index	0.16	0.11	0.19	0.15	−0.04 (0.00)***
Treatment					
Left school at 15 or later	0.79		0.00	1.00	1.00
Instrument					
Born 1933 or later	0.68		0.14	0.83	0.69 (0.00)***
Educational attainment PGI					
1st quintile	0.19		0.23	0.17	−0.06 (0.00)***
2nd quintile	0.20		0.21	0.19	−0.02 (0.01)***
3rd quintile	0.20		0.19	0.20	0.01 (0.05)**
4th quintile	0.21		0.21	0.21	0.00 (0.58)
5th quintile	0.21		0.16	0.22	0.06 (0.00)***
Selected controls					
Female	0.54		0.53	0.55	0.02 (0.01)***
Age	74.24	5.34	77.05	73.49	−3.56 (0.00)***
Birth year	1,935.03	5.24	1,929.80	1,936.42	6.62 (0.00)***
Observations	18,705		3,927	14,778	

Note: Means and standard deviations (continuous variables) for the full estimation sample, as well as means for the treatment ($E_i = 1$) and control ($E_i = 0$) groups. The last column reports the raw difference in means and p-values of a t-test for equality of means. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels. The full set of summary statistics, additionally including parental education, survey wave, and principal components of the genetic data, is reported in Appendix Table A.2.

The outcome Y_{it} is the frailty index of individual i in survey wave t. E_i indicates whether i stayed in school until at least age 15, and $G_i \in \{1, \ldots, 5\}$ denotes i's quintile of the EA PGI. The quintile indicators $\mathbb{1}[G_i = g]$ let average frailty differ across quintiles, and the interactions $E_i \times \mathbb{1}[G_i = g]$ let the effect of schooling vary across them.

Since schooling is endogenous, I instrument each interaction $E_i \times \mathbb{1}[G_i = g]$ with $Z_i \times \mathbb{1}[G_i = g]$, where Z_i indicates birth in 1933 or later, the first cohort affected by the 1947 increase in the minimum school-leaving age to 15. Instrumenting via 2SLS yields one first stage per PGI quintile:

$$E_i \times \mathbb{1}[G_i = g] = \alpha_g^{fs} \mathbb{1}[G_i = g] + \pi_g Z_i \times \mathbb{1}[G_i = g] + \theta_{g1}^{fs} c_i + \theta_{g2}^{fs} Z_i \times c_i + X_{it}' \gamma_g^{fs} + u_{git}, \quad g = 1, \ldots, 5. \quad (3)$$

The coefficient π_g is the jump at the 1933 cutoff in the share of individuals in quintile g who stayed in school until at least age 15, that is, the complier share in that quintile. The 2SLS estimation replaces each interaction $E_i \times \mathbb{1}[G_i = g]$ in Equation 2 with its predicted value from the first stage in Equation 3. Each β_g then identifies the local average treatment effect (LATE) of schooling on frailty for compliers in quintile g. Differences across the coefficients β_g capture how the effect of E_i on Y_{it} varies along the EA PGI distribution and thus reveal patterns of gene-environment interaction.

Both stages include the running variable c_i = birth year – 1933, a linear cohort trend centered at the 1933 cutoff, and its interaction with Z_i. The interaction allows the slope to differ on either side of the cutoff. X_{it} contains indicator variables for gender, age, survey wave, and mother's and father's school-leaving age, as well as the first ten principal components of the genetic data as controls. Standard errors are clustered at the individual level.

3.4 Assumptions and identification

Assumptions. Identification rests on three main assumptions. The first is that the 1947 reform is a valid instrument for education. Instrument relevance is not in doubt: the higher leaving age induced a large share of students to increase their attainment (see Figure 1), and my first stage is correspondingly strong (see Section 4). Instrument exogeneity is also likely to hold, given that eligibility for the compulsory schooling reform depends on date of birth and that the 1944 Education Act came years after these cohorts were born.

Second, I assume that the exclusion restriction holds. This requires that nothing but compulsory schooling shifts at the threshold. The leading concern is exposure to the Second World War. Both Clark and Royer (2013) and Banks and Mazzonna (2012), who also use this reform, conclude that wartime disruptions did not vary systematically across the cutoff. The cohort trends I include absorb gradual variation in such exposure, and my estimates hold for narrower bandwidths (see Section 4.6). The exclusion restriction could also be violated if the reform affected always-takers, for example through spillovers if it changed not only instruction time but also school quality. However, the compliers in this particular reform were almost exclusively students in lower-track schools that ended at the minimum leaving age (Clark 2023). This arguably left little scope for spillovers onto those who would have stayed on regardless. Furthermore, as enrolment expanded, so did school resources, leaving class sizes and teacher qualifications largely unaffected (Clark and Royer 2013). The additional year of schooling did not yield formal qualifications, which would have required remaining in school until at least age 17 (Banks and Mazzonna 2012; Clark 2023). The reform lengthened schooling without changing its character. The third assumption, monotonicity, rules out defiers, that is, individuals who left school earlier only because they were subject to an additional compulsory year. Defiance of this kind seems unlikely.

Attrition. In longitudinal studies of older individuals, such as ELSA, nonrandom attrition may be problematic if it is related to the treatment. If differential attrition is present, those who remain in the sample skew the treatment (or control) group, potentially biasing the estimated treatment effect. For this study, a While the debate on a causal effect of compulsory schooling on mortality is not settled and depends, among other things, on institutional setting (Galama, Lleras-Muney, and van Kippersluis 2018), Clark and Royer (2013) find no effect on mortality from the 1947 compulsory schooling reform. Even if differential survival were present, the direction of the resulting bias is predictable in my application. Since frailty increases strongly with age, longer survival in my treated group would mean that more frail individuals stay alive, shifting the treatment group towards higher frailty. Given that I

find a reduction in frailty due to education, this means I would likely underestimate the true effect. I nevertheless test for panel attrition and find no evidence of it, and I estimate my main specification, reweighted to account for attrition. See Section 4.6 for details.

4 Results

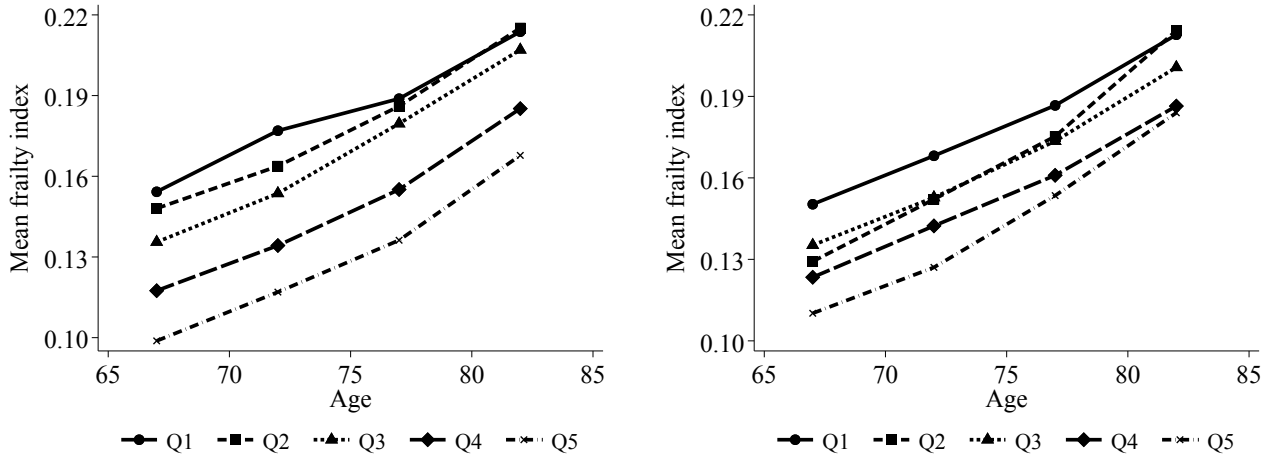
4.1 Descriptive patterns

In this section, I first show that the EA PGI is generally informative about health differences later in life. I then document that the educational health gap in my setting varies with the EA PGI in a way that is already consistent with the gene-environment interaction in the causal effect of education on health that I estimate in the next section.

The EA PGI is designed to predict educational attainment, although, depending on the discovery study and cohort, it currently explains only about 11 to 16 percent of the variation in educational attainment (Lee et al. 2018; Okbay et al. 2022). Nevertheless, genetic achievement gaps in school are as large as socioeconomic achievement gaps and follow the same trajectory across grades (Houmark et al. 2025). The EA PGI also appears to capture information relevant to outcomes beyond schooling. It has a causal effect on occupation, income, and home ownership (Carvalho 2025). It is correlated at the population level with personality (Mõttus et al. 2017), non-cognitive skills (Belsky et al. 2016), early-life physical and mental health (Hughes et al. 2024; Bolyard and Savelyev 2025), and health behaviors such as smoking (Wedow et al. 2018). It further predicts diagnoses of ten cardiometabolic, lung, musculoskeletal, neurological, and psychiatric diseases (Okbay et al. 2022). Taken together, these findings may point to pleiotropy, where the genetic variants in the EA PGI affect multiple outcomes, either directly or through their effect on educational attainment.

Given these propeties of the EA PGI, Figure 3 compares health gaps and trajectories from age 65 to 84 by socioeconomic status (SES) and EA PGI. It shows means of the frailty index for 5-year age bins, separately for quintiles of an SES index (Panel A) and quintiles of the EA PGI (Panel B). I construct the SES index following Houmark et al. (2025), but adapt it to older individuals. It uses the level of education, total household wealth and pension income. Household wealth is broken down by dividing by the square root of household size. For each of the three components, I rank respondents within each wave of data collection and convert the ranks to percentiles. Ranking within wave ensures that the index is unaffected by inflation or by respondents being observed in different waves. I then average each respondent's percentile rank over all waves and take the average of the three percentile ranks to get the SES index. Each component receives equal weight and by averaging, there is one observation per respondent, so that the cut-offs do not depend on how many times each person was observed. Finally, I assign each respondent to a quintile of the index.

Figure 3 allows for two observations. First, the EA PGI itself is informative about late-life health. As with SES, frailty differs clearly across EA PGI quintiles, with those in the highest quintile being the healthiest (lowest frailty) and those in the lowest quintile the frailest. These differences are already present at ages 65 to 69 and widen almost in parallel with age. Second, the frailty gap by EA PGI



A. Socioeconomic differences in old-age frailty

B. EA PGI differences in old-age frailty

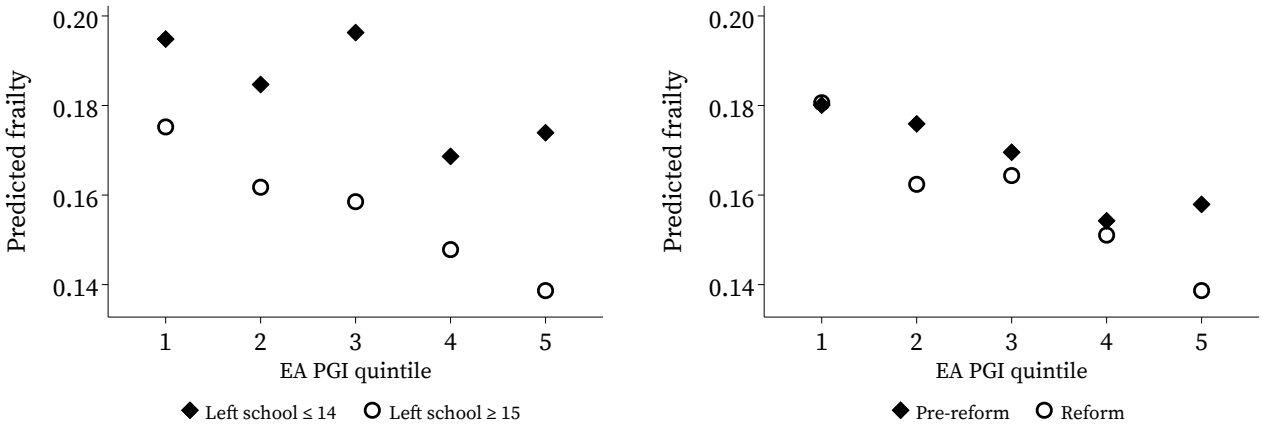
FIGURE 3 – Socioeconomic and genetic frailty gaps

Note: Progression of the average frailty index from age 65 to 84 in 5-year bins and separately by quintile of a socioeconomic score (Panel A) that includes education, household wealth and pension income and quintile of the EA PGI (Panel B). Averages are plotted at the midpoint of each 5-year bin. Sample: 26,850 observations from 6,083 individuals

is almost as large as the SES gap. The total frailty difference by EA PGI (between quintiles 1 and 5) matches the frailty difference between SES quintiles 1 and 4.

Next, I investigate descriptive patterns of gene-education interactions by illustrating that the health differences between more and less educated individuals in my sample vary with the EA PGI. Figure 4 shows predicted frailty, adjusted for age and gender, across quintiles of the EA PGI. Panel A splits the sample by whether individuals left school at age 15 or later, and Panel B by exposure to the 1947 compulsory schooling reform. Panel A reveals three patterns. First, individuals with more schooling are healthier than those with less schooling. Second, frailty decreases in the EA PGI, as documented above. Third, this decrease is steeper for those with more schooling. In this group, individuals at the high end of the EA PGI are considerably less frail than those in quintiles 1–3. Panel B compares groups defined by exposure to the compulsory schooling reform. The same patterns hold, but they are less pronounced. Differences between the two groups are smaller in every PGI quintile. There is little to no difference between the two groups in quintiles 1, 3, and 4, but still a considerable difference in quintiles 2 and 5.

Overall, this hints at potential interactions between the EA PGI and education in late-life frailty: those with more education are less frail, but how much less differs by quintile of the EA PGI and those with more education and in the highest quintile are particularly healthy. These patterns are descriptive and do not yet imply a causal effect of (compulsory) schooling on frailty, nor that such an effect would vary by the EA PGI. Instead, it may be that generally healthier individuals are acquiring more education and that those who are healthier in early life are also healthier in later life. Since both educational attainment and early-life health are correlated with the EA PGI, these descriptive patterns could reflect simple correlations and reverse causality.



A. Average frailty by educational attainment and EA PGI

B. Average frailty by reform exposure and EA PGI

FIGURE 4 – Predicted frailty by educational attainment, reform exposure and EA PGI quintile

Note: Panel A: Predicted frailty index by education indicator and quintile of the EA PGI. Panel B: Predicted frailty index by exposure to the 1947 UK compulsory schooling reform, the instrument. Predictions are non-parametrically adjusted for age and gender.Variable definitions according to section 3.1. Based on the main estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals.

4.2 The causal effect of education on frailty by EA PGI

To estimate a causal effect of education on frailty and investigate whether it varies by the EA PGI, I proceed to estimate a fuzzy regression discontinuity model, as introduced in Section 3.3. I present the results alongside an results from an OLS regression in Table 2. I show both estimation results for an average effect without interacting by EA PGI and an interacted regression that estimates the relationship between my education indicator and the frailty index in each quintile of the EA PGI according to Equation 2.

The OLS regressions in columns (1) and (2) show a statistically significant negative correlation between the treatment E_i (staying in school until at least age 15) and old-age frailty, both in the pooled regression and in each PGI quintile. Across all quintiles of the EA PGI, staying in school for the additional year is associated with a decrease in diagnosed conditions and impairments by about 2 – 4 percentage points. All estimates are relatively small, but they consistently have a negative sign.

The IV estimate in Column (3) of Table 2 indicates that, on average across the EA PGI, an additional year of education reduces frailty by about 3 percentage points, 18 percent of the mean. Column (4) shows that this average is mainly driven by individuals in the highest quintile of the EA PGI, whose frailty decreases by almost 6 percentage points (37 percent of the mean). This is the only statistically significant interaction effect, suggesting that more education reduces the share of diagnosed conditions only among those with a high EA PGI.

The paper's main contribution, however, is to document how the effect of education on old-age frailty changes along the EA PGI. The estimates in Column (4) increase in magnitude with the EA PGI. The effect is practically zero in the lowest quintile. The coefficients for quintiles 2 to 4 are larger but similar to each other, and they are about half the size of the coefficient for the highest quintile. Although

TABLE 2 – The effect of education on old-age frailty by EA PGI

DV: Frailty index	(1) OLS	(2) OLS	(3) IV	(4) IV	(5) RF
E_i	−0.028***		−0.029**		
	(0.005)		(0.014)		
$E_i \times (G_i = 1)$		−0.021**		−0.010	
		(0.010)		(0.016)	
$E_i \times (G_i = 2)$		−0.022**		−0.029	
		(0.009)		(0.018)	
$E_i \times (G_i = 3)$		−0.038***		−0.026	
		(0.010)		(0.020)	
$E_i \times (G_i = 4)$		−0.020**		−0.026	
		(0.008)		(0.019)	
$E_i \times (G_i = 5)$		−0.033***		−0.059**	
		(0.010)		(0.026)	
$Z_i \times (G_i = 1)$					−0.006
					(0.011)
$Z_i \times (G_i = 2)$					−0.016
					(0.010)
$Z_i \times (G_i = 3)$					−0.013
					(0.010)
$Z_i \times (G_i = 4)$					−0.011
					(0.009)
$Z_i \times (G_i = 5)$					−0.022**
					(0.010)

Note: E_i is the education indicator (staying in school to at least age 15), G_i is quintile of the educational attainment PGI. Columns (1) and (2) report OLS estimates, (3) to (5) IV estimates instrumenting E_i the binary instrument Z_i (born 1933 or later), and Column (5) the reduced form where frailty is regressed directly on interactions of Z_i and G_i. Columns (2) and (4) report the full effect of E_i within each PGI quintile as given by Equation 2. All specifications control for gender, survey wave, age, and mother's and father's age when leaving full-time education. IV specifications add the running variable (re-centered birth year) and its interaction with the instrument. Interacted specifications also include the first ten genetic principal components. Estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

these middle coefficients are not statistically significant, they are potentially economically relevant. On average, moving up one PGI quintile increases the effect of education on old-age frailty by 1.2 percentage points of the frailty index.[1] This pattern is consistent with a gene-education complementarity, where those who have more genetic variants strongly correlated with educational attainment and learning benefit more from increased instruction time.

Since a LATE corresponds to the ratio between reduced form and first stage, a large coefficient could in principle be driven by a small first stage. The first stage, reported in Table A.3 does indeed decrease in EA PGI, but remains large, even in the highest quintile, with about 36 percent being compliers. Furthermore, the reduced-form estimates in Column (5), where frailty is regressed directly on interactions between the instrument and EA PGI quintiles, suggest the same pattern as the IV results. There is a statistically significant effect of eligibility for the compulsory schooling reform on

[1]This is the average change between adjacent quintiles, $(\beta_5 - \beta_1)/4 = (-0.059 - (-0.010))/4 \approx -0.012$.

frailty only for those in the highest PGI quintile. The coefficient increases in EA PGI and is about twice as large in quintile five.

4.3 Discussion

My main result, that the effect of education on old-age frailty increases in the EA PGI, mirrors findings for other outcomes. The effects of education on income and on cognitive ability in old age both increase in the EA PGI, driven by individuals at the top of the EA PGI distribution (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026). I find that the effect of education on frailty rises by 1.2 percentage points per EA PGI quintile on average, compared to a large increase of 0.46 words per quintile for word recall (Hollenbach, Schmitz, and Westphal 2026). The effect on frailty, however, increases relatively smoothly in the EA PGI, while the effect on cognition evolves non-linearly. Studies of other health outcomes also find that the gains from education vary with genetic endowments, but they measure these endowments with PGIs for the specific condition rather than with the EA PGI. The additional education from the 1972 UK schooling reform reduced Alzheimer's diagnoses, particularly for those at high genetic risk of the disease (Barcellos et al. 2025). It also improved body size, lung function, and blood pressure, mostly for those at the highest genetic risk of overweight (Barcellos, Carvalho, and Turley 2018). This study finds no differences by the EA PGI, but it measures health in middle age, whereas I study old age.

No study estimates the causal effect of education on old-age frailty with a comparable design. Huang et al. (2024) construct a Wald estimate from genetic discovery studies. For a set of genetic variants that predict schooling, they divide the effect of each variant on frailty (the "reduced form") by its effect on schooling (the "first stage") and obtain a LATE. Their estimate is large: one SD more schooling reduces frailty by about half an SD. My pooled estimate of the effect of schooling on frailty is roughly as large as the effect of retirement on the frailty index that Abeliansky and Strulik (2023) find for women. My estimate for the fifth EA PGI quintile (about 6 percentage points) exceeds their largest effect, a 4.5 percentage point reduction in frailty from retiring for men with low education.

More broadly, the evidence on the causal effect of education on later-life health in the UK is mixed. Clark and Royer (2013) estimate very small reduced-form effects of the 1947 and 1972 compulsory schooling reforms on self-rated health, and Jürges, Kruk, and Reinhold (2013) find no clear effects on self-rated health or blood-based markers of inflammation. In contrast, Silles (2009) reports sizable 2SLS estimates showing that education increases self-reports of good health by 6.5 percent, of having no long-term illness by 7.5 percent, and of having no activity-limiting illness by 5 percent. Using UK Biobank data and the 1972 reform, Davies et al. (2018) show that additional education reduced hypertension by 8 percent, diabetes by 3.6 percent, and strokes and heart attacks by about 1 percent each. It also lowered the 90th percentile of body size by 0.4 SD and the obesity rate by 8 percentage points, and raised the 30th percentile of blood pressure by a third of an SD (Barcellos, Carvalho, and Turley 2023). All of these studies observe individuals at younger ages than I do and focus on single health outcomes.

4.4 Distributional effects on frailty

An effect on the frailty index does not reveal where in the distribution it occurs. Where education reduces frailty matters for interpretation, because the frailest individuals in particular have lower quality of life, higher care needs, and lower chances of survival (Song, Mitnitski, and Rockwood 2010). I therefore split the frailty index into three commonly used categories (see, e.g., Song, Mitnitski, and Rockwood 2010; Davies et al. 2021): robust (frailty ≤ 0.08), pre-frail ($0.08 <$ frailty < 0.25), and frail (frailty ≥ 0.25), and use them as outcomes in my main IV model (Column 4 of Table 2). A lower probability of being frail would imply that individuals avoid the worst consequences of frailty, whereas a higher probability of being robust would mean more individuals thriving in old age.

Table 3 reports how the additional education shifts individuals between categories. By construction, a decrease in one category implies an increase in one or both of the others, and vice versa. Overall, the additional education decreases the probability of being frail. The decrease is especially large in the highest EA PGI quintile, where it is accompanied by sizable but statistically insignificant increases in the probability of being robust (the lower end of the distribution) and of being pre-frail (the middle). In quintiles 2 and 3, the probability of being frail also decreases, but individuals move into the pre-frail rather than the robust category, which may partly explain the lack of a statistically significant effect on the overall index in these quintiles since for these individuals, the decrease was likely smaller than for those in the highest quintile.

The results suggest that the additional schooling reduced frailty at the upper end of its distribution, consistent with individuals avoiding or delaying the accumulation of a debilitating number of deficits. In the highest EA PGI quintile, it may also have kept individuals robust, that is, nearly deficit-free, for longer.

4.5 Marginal treatment effects of education on frailty

Standard IV estimates of interaction effects do not necessarily reflect the true interaction effect. The estimator cannot separate changes in the treatment effect along the interaction variable (the interaction effect) from changes in the characteristics of compliers along the interaction variable (Hollenbach, Schmitz, and Westphal 2026). Hollenbach, Schmitz, and Westphal (2026) formalize two conditions that, if met, can cause 2SLS estimates to differ in size from the true interaction effect. First, compliers must differ in unobserved characteristics across strata of the interaction variable. Second, treatment effects must exhibit essential heterogeneity, meaning that latent resistance to treatment is correlated with the individual treatment effect, so individuals select into treatment based on unobserved gains (or losses) (see, e.g., Heckman, Urzua, and Vytlacil 2006). If both conditions apply, the 2SLS interaction estimate compares treatment effects of compliers located at different points of latent resistance to treatment. Since resistance is tied to the size of the treatment effect, the resulting difference in LATEs may differ from the true interaction effect, which would entail a comparison of compliers at a fixed point of latent resistance. The latter can be obtained by estimating marginal treatment effects (MTEs).

TABLE 3 – IV results at different margins of the frailty index

Dependent variable	Robust	Pre-frail	Frail
$E_i \times (G_i = 1)$	−0.014	0.072	−0.058
	(0.047)	(0.054)	(0.053)
$E_i \times (G_i = 2)$	−0.019	0.124**	−0.105*
	(0.055)	(0.063)	(0.059)
$E_i \times (G_i = 3)$	−0.016	0.134*	−0.118*
	(0.064)	(0.072)	(0.065)
$E_i \times (G_i = 4)$	0.060	0.009	−0.069
	(0.065)	(0.072)	(0.065)
$E_i \times (G_i = 5)$	0.103	0.079	−0.181**
	(0.091)	(0.097)	(0.083)
Observations	18,705	18,705	18,705
Individuals	3,415	3,415	3,415

Note: Dependent variables are indicators for being robust (frailty index ≤ 0.10), pre-frail (0.10 < frailty index ≤ 0.25), and frail (frailty index > 0.25). E_i is staying in school to at least age 15. IV estimates instrumenting E_i with exposure to the compulsory schooling reform (born 1933 or later). All specifications control for gender, survey wave, age, the running variable (re-centered birth year) and its interaction with the instrument. Estimation sample as described in Section 3.2. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The first condition clearly applies in my setting. Appendix Figure A.1 shows the first-stage estimates by EA PGI quintile, which correspond to the share of compliers in each quintile (Appendix Table A.3 reports the exact estimates). The complier share declines monotonically along the EA PGI, from about 65 percent in the lowest quintile to 36 percent in the highest, which is still a sizable share and shows that the 1947 reform had a broad impact. Similar differences in the response to schooling reforms by genetic endowment have been documented by Barcellos, Carvalho, and Turley (2018) for the 1972 UK reform and by Hollenbach, Schmitz, and Westphal (2026) for the 1947 reform. They indicate that the EA PGI enters the cost-benefit considerations behind the decision to stay in school (Hollenbach, Schmitz, and Westphal 2026), as in the choice model underlying the MTE framework (see Appendix Section B.1). This is in line with how the EA PGI is generally seen in the schooling context where students with a higher EA PGI acquire skills more effectively and likely face lower costs of schooling (Houmark, Ronda, and Rosholm 2024; Carvalho 2025). Since individuals at different points of the EA PGI react differently to the reform, the compliers who identify the interaction effect likely also differ in the unobserved part of their school choice, which reflects, for example, tastes or preferences. If this unobserved part is correlated with the returns to education, the 2SLS interaction effect compares returns of compliers who are not comparable and may therefore be skewed.

This second part, essential heterogeneity, has been established as a feature of educational choices in the form of selection on gains (Carneiro, Heckman, and Vytlacil 2011; Kamhöfer, Schmitz, and Westphal 2019; Westphal, Kamhöfer, and Schmitz 2022; Hollenbach, Schmitz, and Westphal 2026). Nevertheless, as Kowalski (2023) points out, even within the same (natural) experiment, patterns of selection into treatment can differ substantially depending on the outcome. In fact, Barcellos,

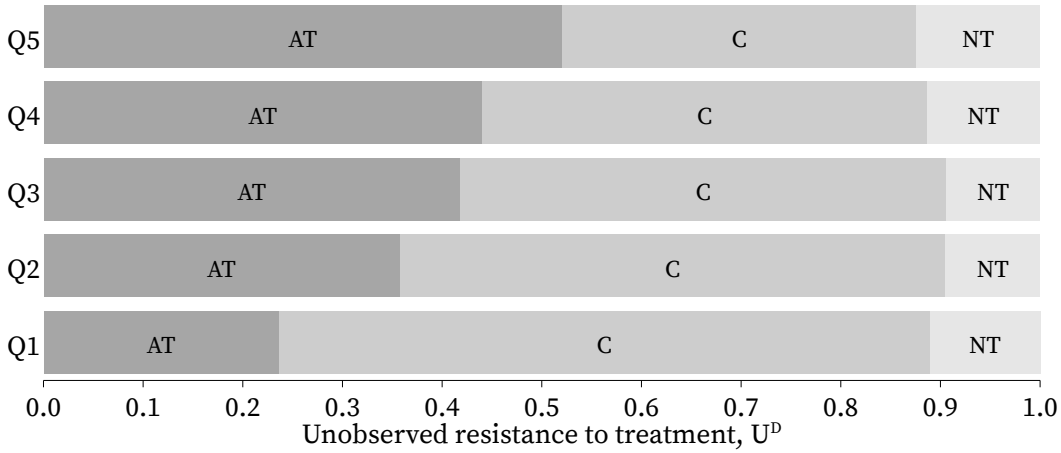


FIGURE 5 – Location of always-takers, compliers, and never-takers along unobserved resistance to schooling, by EA PGI quintile

Note: Shares of always-takers (AT), compliers (C), and never-takers (NT) in each quintile of the EA PGI, ordered along the unobserved resistance to taking more schooling, U_i^D. Low values indicate a high willingness to stay in school beyond age 14. The share of always-takers is the predicted share in school until at least age 15 among those not exposed to the reform, and the complier share is the first-stage estimate from Equation 3. Shares are estimated with a linear probability model and evaluated at the cutoff. Estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals.

Carvalho, and Turley (2021) do not find evidence of selection on gains in wage returns to the 1972 UK schooling reform. Whether 2SLS recovers the true interaction effect in my setting is therefore an empirical question, which I address by estimating MTEs by EA PGI quintile.

Unobserved resistance and response types. The MTE framework orders individuals by their unobserved resistance to schooling, U_i^D (Heckman and Vytlacil 2005). It captures everything that makes staying in school beyond age 14 more or less attractive to an individual but is not observed in the data, such as preferences, perceived costs, or a distaste for school. It is expressed as a rank between 0 and 1 within each EA PGI quintile. For example, an individual with $U_i^D = 0.2$ is more reluctant to stay in school than 20 percent of the individuals in their quintile and less reluctant than the remaining 80 percent. Like complier status, U_i^D is not observed at the individual level. This ordering is directly related to the three response types. A lways-takers have the lowest resistance stay in school and do so even without the reform. They are located at the left end of the unit interval. Individuals with somewhat higher resistance stay in school only because the reform requires them to. These are the compliers. Finally, individuals with the highest resistance leave school early even though the reform requires them to stay. These never-takers are located at the right end of the unit interval. Clark and Royer (2013) report that there were individuals who did not comply with the schooling reform and they consisted mainly of children born in the summer months who turned 15 in between school years and were therefore allowed to leave.

Figure 5 shows the shares of each type by EA PGI quintile ordered on the U_i^D interval. The width of each segment corresponds to the share of the respective type. The decline in the complier share is

the first-stage pattern discussed above. The figure additionally shows that it is mirrored by a steady increase in the share of always-takers, from 24 percent in the lowest quintile to 52 percent in the highest, while the share of never-takers remains between 9 and 12 percent throughout. As a result, compliers in the lowest quintile make up a large part of the unit interval, from $U^D = 0.24$ to 0.89, while those in the highest quintile are concentrated at higher levels of resistance, between 0.52 and 0.88 since most of them are more eager to stay in school longer. The increasing share of always-takers shows that the EA PGI contains signals that are informative about choice behavior in this specific educational decision.

Marginal treatment effects. The MTE is the effect of the additional year of schooling for individuals at a given point of resistance u in quintile g, $\text{MTE}_g(u) = \mathbb{E}[Y_i^1 - Y_i^0 \mid U_i^D = u, G_i = g]$, where Y_i^1 and Y_i^0 denote frailty with and without the schooling treatment. Plotted along U_i^D, it shows how the effect of schooling changes with resistance to schooling. The MTE consists of two parts, the marginal treatment responses (MTRs): the average frailty of treated individuals at resistance u, $\mathbb{E}[Y_i^1 \mid U_i^D = u, G_i = g]$, and without treatment, $\mathbb{E}[Y_i^0 \mid U_i^D = u, G_i = g]$. The MTE can be used to calculate different estimands, among them the LATE. I can therefore obtain an estimate similar to the β_g in my 2SLS estimation, average of the MTE over the compliers of quintile g. An interaction effect can be obtained by investigating differences between the G_i-specifc MTE curves.

Estimation. With a binary instrument, the data identify four average outcomes in each quintile: the treated frailty of always-takers and compliers and the untreated frailty of compliers and never-takers (Imbens and Rubin 1997). These are averages over the corresponding intervals of U^D, and the MTR curves through them cannot be point-identified without further assumptions. A common solution is to assume that the MTRs are linear and to draw a line through the midpoint of the treated average for AT and the treated average for C (yielding the treated MTR) and through the midpoints of the untreated averages for C and NT, yielding the untreated MTR (Brinch, Mogstad, and Wiswall 2017). Instead, I follow Mogstad, Santos, and Torgovitsky (2018) and Hollenbach, Schmitz, and Westphal (2026) and estimate bounds on the MTEs.

To do so, I approximate each MTR by a flexible curve on the unit interval, a Bernstein polynomial of degree six, and obtain the MTE of each quintile as the difference between its treated and untreated MTR. Many such curves are consistent with the four average outcomes in a quintile. Restrictions on their shape, described below, narrow down this set. Among the remaining curves, I search for those that yield the smallest and the largest value of the quantity of interest. For the effect within a quintile, it is the average MTE over the interval $U_i^D \in [0.55, 0.85]$, the counterpart of β_g in the 2SLS estimation. For the G×E interaction, it is the difference between this average in the highest and in the lowest quintile. I optimize this difference directly rather than subtracting the quintile-specific bounds from each other (Appendix Section C). The main strength of this approach is that it does not require linearity, which is a strong assumption, and instead relies on much weaker shape restrictions while remaining grounded in the data. The drawback is that the bounds are not always informative.

The interval $[0.55, 0.85]$ is central to the comparison across quintiles. As in Hollenbach, Schmitz, and Westphal (2026), I choose it because it is the range of resistance in which compliers are found in every quintile (Figure 5). In the highest quintile, it almost coincides with the complier range, while in the lowest quintile, compliers extend much further toward low resistance. Averaging over this common interval has two advantages. First, the quintile-specific effects still refer to compliers, so they are comparable to the 2SLS estimates. Second, the interaction effect compares individuals with the same resistance to schooling in the highest and the lowest quintile, so it is not affected by the differences in complier composition discussed above. Outside this interval, the MTE curves depend more strongly on the restrictions, as I discuss below.

The restrictions on the flexible curves I fit are as follows. First, all values of every MTR lie between 0 and 1, the range of the frailty index. Second, when averaged over the appropriate share of always-takers, compliers, or never-takers, the MTRs reproduce the four average outcomes in each quintile, up to a small tolerance that accounts for sampling noise in these averages. Third, both MTRs are non-decreasing in U_i^D. Fourth, the MTEs are non-increasing in U_i^D. Both of these monotonicity restrictions are weak, meaning that the curves can also be flat. A flat MTE, that is, no essential heterogeneity, is therefore permitted. I base the third and fourth restriction on a test for essential heterogeneity with linear MTRs (Brinch, Mogstad, and Wiswall 2017), reported in Appendix Section B.3. Both linear MTRs have positive slopes, and since the untreated MTR increases more steeply, the implied MTE slope is negative, although none of these slopes is statistically significant, which is why I allow for flat curves as well. Fifth, the MTRs have similar shapes in every quintile, so that the EA PGI mainly shifts their level. As Hollenbach, Schmitz, and Westphal (2026) mention, this additive separability assumption is commonly used to estimate MTEs and implicitly made in all 2SLS estimations. I obtain standard errors from a Bayesian bootstrap with 300 replications, clustered at the individual level. Appendix Section B provides a more detailed introduction to MTEs and Appendix Section C describes each estimation steps.

Results. Table 4 reports bounds on the average MTE effect of education on frailty by quintile of the EA PGI as well as the interaction effect. The bounds are obtained optimizing over the interval $[0.55, 0.85]$ which fixes unobserved resistance to schooling for each MTE, while remaining comparable to the 2SLS results above. Panel A shows that both bounds are negative in every quintile, suggesting that the additional year of schooling reduced frailty along the entire EA PGI distribution. As in the 2SLS results, the effect is largest in the highest quintile, at 5.4 to 5.8 percentage points, about twice as large as in the other quintiles, and the bounds there are very close to the 2SLS estimate of 5.9 percentage points (Column 4 of Table 2).

Statistical significance depends on the minimum, the bound closest to zero. If the smallest effect consistent with the data and the restrictions differs significantly from zero, so does the unknown point effect. A significant maximum by itself cannot rule out a zero effect. Accordingly, the effect in the highest PGI quintile is significant at the 1 percent level. In the lowest quintile, the bounds range from 2.0 to 3.1 percentage points, two to three times the 2SLS estimate of 1.0 percentage point, but are not

TABLE 4 – Bounds on the effect of education on frailty for compliers on the common margin

DV: Frailty index	(1) Min	(2) Max
Panel A: Average MTE on $U^D \in [0.55, 0.85]$		
$E_i \times (G_i = 1)$	−0.020	−0.031**
	(0.017)	(0.019)
$E_i \times (G_i = 2)$	−0.034**	−0.042**
	(0.018)	(0.020)
$E_i \times (G_i = 3)$	−0.025	−0.033*
	(0.022)	(0.023)
$E_i \times (G_i = 4)$	−0.025*	−0.030*
	(0.018)	(0.020)
$E_i \times (G_i = 5)$	−0.054***	−0.058***
	(0.021)	(0.023)
Panel B: Interaction effect on $U^D \in [0.55, 0.85]$		
$E_i \times (G_i = 5) - E_i \times (G_i = 1)$	−0.024	−0.035*
	(0.024)	(0.025)

Note: Minimum and maximum of the average marginal treatment effect of staying in school until at least age 15 (E_i) on the frailty index over $U^D \in [0.55, 0.85]$, separately for each quintile of the EA PGI (G_i, Panel A), and on the difference between the highest and the lowest quintile (Panel B), obtained by minimizing and maximizing the respective effect over all curves consistent with the data and the restrictions. Min and Max refer to the size of the effect: Min is the bound closest to zero, Max the bound furthest from zero. The interaction effect is optimized directly and is not the difference of the bounds in Panel A. Restrictions: MTRs bounded by $[0, 1]$ and non-decreasing in U^D, MTEs non-increasing in U^D, similar shapes across quintiles, and must reproduce average outcomes from the data. Type shares and average outcomes control for the running variable and its interaction with the instrument, gender, survey wave, age, mother's and father's age when leaving full-time education, and the first ten genetic principal components. Estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals. Standard errors from a Bayesian bootstrap with 300 replications, clustered at the individual level, in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

statistically significant. This may hint at the 2SLS estimate in the lowest quintile skewing downwards since compliers there range from $U_i^D = 0.24$ to 0.89, many of whom lie outside the comparable interval and, according to Figure 6, benefit less from the additional year of schooling. In quintiles 2 to 4, the bounds lie between 2.5 and 4.2 percentage points, which is slightly larger than the 2SLS estimate. The effect is significant at the 5 percent level in the second quintile, only at the 10 percent level in the fourth, and not at all in the third.

Since the MTE estimate of the effect of education on frailty in the lowest PGI quintile is larger than the 2SLS estimate, the interaction effect in Panel B is smaller than its 2SLS counterpart. The effect of schooling on frailty is 2.4 to 3.5 percentage points larger in magnitude in the top PGI quintile than in the bottom. Both bounds are negative, implying likely larger reduction in frailty in the highest quintile. The bounds generally support the a reinforcing G×E interaction, but are imprecisely estimated. The

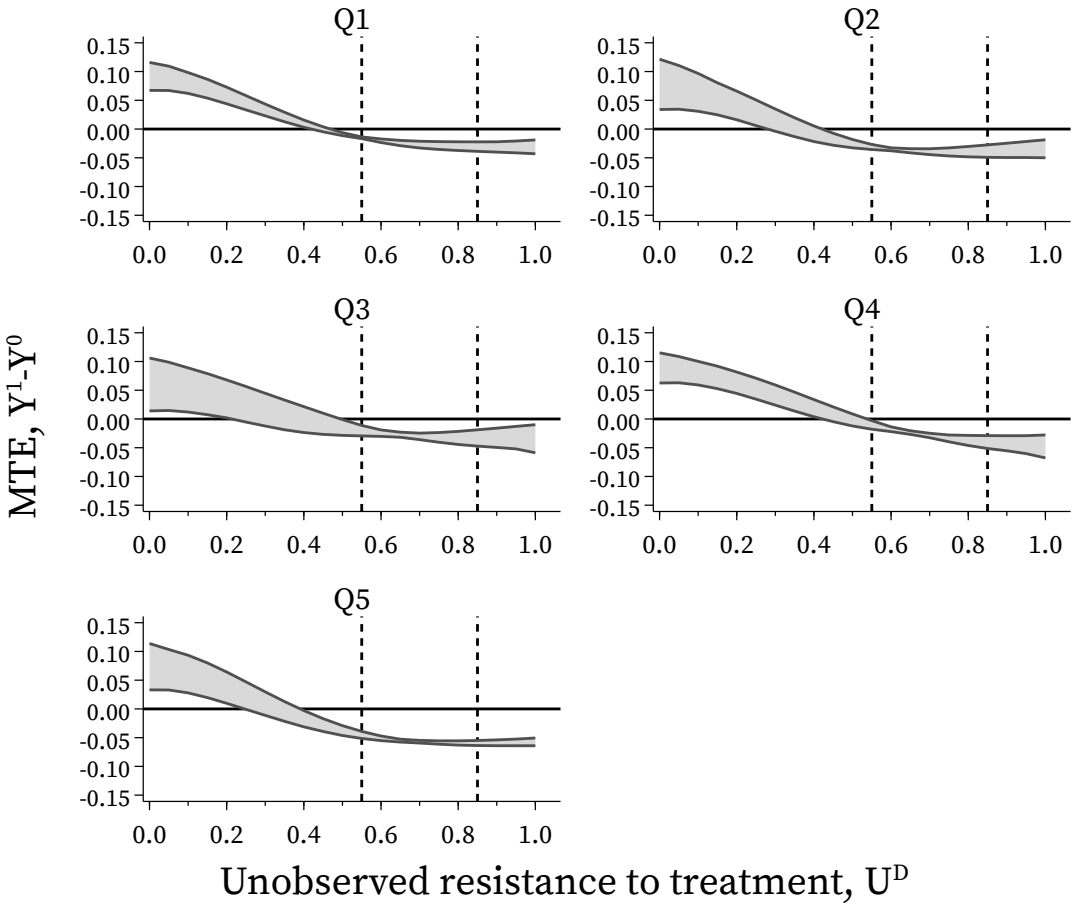


FIGURE 6 – Marginal treatment effects of education on frailty by EA PGI quintile

Note: Bounds on the marginal treatment effect of staying in school until at least age 15 on the frailty index, $\mathbb{E}[Y_i^1 - Y_i^0 \mid U_i^D = u, G_i = g]$, along the unobserved resistance to schooling U^D, separately for each quintile of the EA PGI. The grey area shows, at each value of U^D, the smallest and largest MTE consistent with the data and the restrictions described in Section 4.5. Dashed vertical lines mark the interval $[0.55, 0.85]$, which is populated by compliers in all quintiles. MTRs are approximated by Bernstein polynomials of degree six. Estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals. Appendix Section C describes the estimation in detail.

interaction effect is smaller than the 2SLS difference of 4.9 percentage points between the highest and lowest quintile, which is driven by a higher effect in the lowest quintile discussed above.

While Table 4 focuses on the common complier range, Figure 6 shows the bounds on the full MTE curves for each quintile and thus how the effect changes with resistance to schooling. Within $[0.55, 0.85]$ and further to the right, the bounds are narrow, negative, and relatively flat. At lower values of U_i^D, where always-takers and some compliers are located, they are wide, positive, and decreasing. Outside the complier range, the effects, and therefore the curves, rely more the shape of the curves, since the unobservable counterfactuals for always and never-takers are based on extrapolations.

The positive values at the left would imply that additional schooling increased frailty for always-takers, which would be unexpected for those most eager to stay in school. More likely, the additional year did

little for the frailty of always-takers. The negative slope of the MTE is mainly driven by the difference in untreated frailty between compliers and never-takers (Appendix Table B.1), and the restriction of a non-increasing MTE carries it over to the left tail. This is why I focus on effects within the interval populated by compliers from all quintiles.

Since lower frailty means better health, the negative MTE slope implies selection into losses where those most willing to stay in school benefit least in terms of frailty. This is surprising, but the pattern is not very pronounced. At the same time, selection on gains would be just as surprising for an outcome like old-age health. Selection on gains describes individuals who choose more education because they anticipate, at least partly, their own returns (Willis and Rosen 1979; Heckman and Vytlacil 2005). This is more straightforward for outcomes such as earnings (Carneiro, Heckman, and Vytlacil 2011; Nybom 2017) and cognitive ability (Hollenbach, Schmitz, and Westphal 2026), which are closely tied to learning and to the payoffs students or their parents would consider, than for health.

In conclusion, the first stage differs strongly by EA PGI, likely because the EA PGI affects the costs of and returns to education and therefore educational choices. I find suggestive evidence that these differences in compliance behavior have affected the 2SLS estimate of schooling on frailty in the lowest EA PGI quintile. The true estimate is likely higher, suggesting that the effect gradient along the EA PGI is less steep. Nevertheless, the MTE estimates match the 2SLS estimate in the highest quintile well, where I find the largest health returns to schooling.

Essential heterogeneity in frailty, here in the form of selection into losses, is not very pronounced and mostly driven by never-takers. Comparing compliers at the same resistance to schooling confirms the main pattern: schooling reduces frailty most strongly in the highest quintile of the EA PGI, with an interaction effect pointing towards EA PGI and compulsory schooling reinforcing eacho other, albeit less precisely estimated as with 2SLS due to the bounding approach.

4.6 Robustness

I investigate the robustness of my main 2SLS results with respect to the bandwidth, age composition, linearity of the cohort trends, and the choice of quintiles, the split for the EA PGI, account for attrition probability by reweighting, and investigate an alternative outcome related to functional aging without the mortality aspect. Table 5 reports the results.

Lowering the bandwidth to 8 and 6 years does not change the nature of the results. Only the coefficient for the highest quintile shows a statistically significant effect of a year of education on frailty. In Column (3), I lower the age threshold from 65 to 50. Because of the birth cohort restriction, this adds only 89 individuals. They still contribute many observations because the birth cohort restriction excludes younger respondents who joined ELSA later. Therefore, those who are added here entered early in the study and are observed over many waves. Adding these observations does not change the results.

Next, I use quartiles of the EA PGI instead of quintiles to estimate the interaction terms. The results show a coefficient close to zero for the lowest quartile of the EA PGI distribution and, again, a sizable

reduction in frailty for the highest quartile. As expected, this effect is smaller than the one for the top 20 %. In this specification, there is also a statistically significant reduction in frailty of 0.04 from the additional year of education for the second quartile.

As discussed above in Section 3.4, attrition in old-age samples may interfere with treatment effects. To test for differential attrition, I first fill in missing waves for all individuals (after the first wave in which they are observed) up to wave 10, keeping only the observations in which the individual would have been within my age range of 65 to 85, so that aging out of the sample is not counted as attrition. I then create an attrition indicator equal to one if the individual did not respond in a given wave and use it as the outcome in two specifications: the pooled 2SLS estimation (without PGI interactions) and a reduced-form regression of attrition directly on the instrument and controls. Appendix Table A.4 reports the results. Both coefficients are close to zero and no statistically significant. So, the reform, and schooling induced by the reform, are unrelated to panel attrition in my sample.

Nevertheless, I also estimate the main regression with inverse probability weights that reweight for panel response (Table 5, Column 7). I obtain the individual-level weights from a logit regression of a response indicator (1 – *attrition*, as defined above) on observables, including the reform dummy, the education dummy, the EA PGI, birth cohort, a wave indicator, as well as age and frailty measured in the individual's first observed wave. The sample for this regression includes all waves after an individual's first appearance until wave 10, as long as their projected age still falls into the 65 to 85 window. Everyone's first wave is excluded, since the inclusion probability is one there. The estimate in Column (7) is my main regression, but weighted by the inverse of the predicted response probability from this logit regression. Again, the results are very similar to my main set of results.

Furthermore, I estimate a specification using only the first observation of each individual. By definition, there can be no panel attrition, since it only uses cross-sectional information. The results do not differ substantially from the main results.

Finally, I estimate the main regression for an alternative outcome, a sub-index of the frailty index that omits severe conditions. These omitted items are cardiovascular, metabolic, cognitive, and other severe diagnoses. The remaining 40 items include impairments in activities of daily living, self-assessments of physical and mental health, and cognitive tests, which cover functional and sensory aspects of the frailty index, but do not relate as much to the mortality dimension. The results are qualitatively and quantitatively (as a share of the standard deviation) comparable to the main results. This shows that the results hold for an alternative outcome that tracks mortality less closely, but instead measures quality of life and autonomy in old age.

4.7 Suggestive evidence on channels

Next, I turn to non-health outcomes that relate to potential channels of an effect of education on old-age health are affected by the reform. I investigate outcomes in three categories: income and pensions (Panel A of Table 6), employment history (Panel B), and health behavior (Panel C). The table also includes mean values of these outcomes and the number of observations and individuals. Each

TABLE 5 – Robustness

DV: Frailty index	(1) Main	(2) BW 8	(3) BW 6	(4) 50–85	(5) Quartiles	(6) IPW	(7) First obs.	(8) Functional
$E_i \times (G_i = 1)$	−0.010	−0.008	−0.017	−0.012	−0.009	−0.009	−0.007	−0.011
	(0.016)	(0.018)	(0.021)	(0.016)	(0.015)	(0.019)	(0.020)	(0.019)
$E_i \times (G_i = 2)$	−0.029	−0.027	−0.035	−0.030*	−0.041**	−0.030	−0.030	−0.027
	(0.018)	(0.020)	(0.022)	(0.018)	(0.018)	(0.021)	(0.023)	(0.022)
$E_i \times (G_i = 3)$	−0.026	−0.029	−0.029	−0.024	−0.019	−0.034	−0.032	−0.029
	(0.020)	(0.021)	(0.024)	(0.020)	(0.017)	(0.022)	(0.025)	(0.024)
$E_i \times (G_i = 4)$	−0.026	−0.028	−0.035	−0.025	−0.052**	−0.030	−0.018	−0.024
	(0.019)	(0.021)	(0.024)	(0.019)	(0.024)	(0.021)	(0.028)	(0.023)
$E_i \times (G_i = 5)$	−0.059**	−0.064**	−0.076**	−0.056**		−0.070**	−0.063*	−0.074**
	(0.026)	(0.030)	(0.035)	(0.026)		(0.029)	(0.033)	(0.032)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,705	16,057	12,962	21,083	18,705	18,705	3,415	18,705
Individuals	3,415	2,844	2,230	3,504	3,415	3,415	3,415	3,415

Note: The dependent variable is the frailty index. E_i is the education indicator (stayed in school until at least age 15) and G_i denotes quintiles of the educational attainment PGI. All columns report IV estimates of the interaction terms of the education indicator with each PGI group, instrumenting education with exposure to the 1947 compulsory schooling reform (born in 1933 or later). Column (1) reproduces the main IV estimates reported in Table 2. Columns (2) and (3) narrow the bandwidth around the reform cutoff to eight and six birth cohorts on either side. Column (4) widens the age range of the estimation sample to ages 50 to 85. Column (5) uses only the first observation per individual. Column (6) weights observations by the inverse of the estimated probability of being observed in a given wave, obtained from a logit model of panel response on baseline characteristics. Column (7) uses an alternative dependent variable, a frailty index constructed from the functional items only. Column (8) splits the PGI into quartiles instead of quintiles, so that only four interaction terms are estimated. All regressions include indicator variables for gender, survey wave, age, and mother's and father's age when they left full-time education, the running variable (re-centered birth year) and its interaction with the instrument, as well as the first ten principal components of the genetic data. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

line is a separate IV regression of the respective outcome on my education indicator instrumented by the reform dummy without PGI interactions using the same sample limitations as the main analysis (10 birth cohorts to either side of the 1933 cutoff and observations from individuals aged 65–85).

The youngest respondents in my sample are 65 years old, so right at the official retirement age. Many have been retired since the first time they were observed in ELSA. Therefore, data on current jobs or earnings reflects jobs respondents have taken on in retirement, rather than the career these individuals have had. I deal with this issue in three ways. First, I use information from ELSA's life history module. There, respondents are asked about their careers and the earnings of their final job in their main career. Second, I look at subjective financial situation, that is, whether a person has ever experienced financial hardship. Third, I use information about the size of public and employer pensions as well as information about receiving a disability pension. The more a person has paid into public or private pension schemes, the more they should be able to take out, giving me a rough idea on whether income might have been affected.

Panel A suggests that the additional education from the 1947 schooling reform did not affect income or pensions. Coefficients on pay in final job, financial hardship, public and employer pensions are close to zero and not statistically significant. Overall, this seems consistent with the small to zero returns from this reform found by Devereux and Hart (2010) or Clark (2023). However, there is a sizable

decrease in the probability of receiving a disability pension. Abeliansky and Strulik (2023) show an occupational gradient in the frailty index. Workers in physically more demanding jobs accumulate health deficits faster. A decrease in the probability of relying on disability pension may hint at less burden and injuries through the job. Unfortunately, ELSA does not provide detailed information about the nature of the jobs in the life history module, like occupational classifiers. Therefore, I am unable to investigate patterns of physical or mental strain of the former jobs.

In terms of employment history, I am able to identify instances of unemployment, self-employment, part-time emplyment, and the number of jobs a person has held. I also investigate the impact of education on taking early retirement, identified from the main sample. The results suggest that there is no effect of the additional year of compulsory education on whether a respondent was ever unemployed, self-employed, ever in part time or on the number of jobs they have had during their working life or has retired early. The literature broadly finds the same: Clark (2023) investigates the effect of the 1947 compulsory schooling reform on several labor market outcomes, including labor force participation, unemployment, selfemployment and finds no effects.

Panel C investigates different aspects of (risky) health behaviors. I observe a large decrease in the probability of ever having smoked and a statistically insignificant, but still sizable estimate suggesting a reduction in the probability of currently smoking. However, there does not seem to be an effect on current alcohol consumption or heavy (risky) drinking, which, according to official UK advisory, is more than 14 units per week (UK Department of Health and Social Care 2016). In terms of physical activity, there is no effect of education on doing any light activity, but sizable positive effects on doing moderate and vigorous activity. Nevertheless, this does not seem to translate to weight reductions, as the estimate on BMI is not statistically significant and not of relevant size given the high average.

Better health behavior may be able to explain reductions in old-age frailty from increased education. Strulik (2018a) uses a health deficit model centered around frailty and calibrated to US data to show that more educated inviduals delay the accumulation of deficits through lower unhealthy consumption (e.g. smoking) and increased health investments (e.g. exercise). In Strulik (2018b), he shows that smoking drives long-run health trajectories and finds longevity gains from not smoking. Lack of exercise has long been identified as a key risk factor for old-age morbidity (World Health Organization 2009). Futhermore, evidence from RCTs randomizing incentives for exercise and exercise itself show that increased physical activity reduces blood sugar in middle-aged diabetics and helps manage the disease (Aggarwal, Dizon-Ross, and Zucker 2026) and increases functional health in old age (Pahor et al. 2014). Furthermore, both increased exercise and reduced smoking are frequently reported as key mechanisms for the effect of retirement on old-age health (see, e.g. Insler 2014 or Eibich 2015).

4.8 Conclusion

In this study, I examine how the effect of education on physical health in old age varies with genetic propensity for educational attainment, using data on individuals aged 65–85 from the English Longitudinal Study of Ageing. First, I document a sizable genetic gap in old-age frailty. The difference in frailty between the lowest and the highest quintile of the EA PGI is about as large as the difference

TABLE 6 – Effect of education on non-health outcomes

	IV estimate	Mean	Observations	Individuals
A. Income and pensions				
Log final pay in last job	−0.117	16,759	2,118	2,118
	(0.236)			
Ever severe financial hardship	−0.024	0.177	3,133	3,133
	(0.055)			
Log public old-age pension income	0.042	5,704	30,941	6,711
	(0.150)			
Log private/employer pension income	0.749	4,343	30,940	6,697
	(0.719)			
Receives disability pension	−0.066**	0.137	31,251	6,716
	(0.032)			
B. Employment history				
Ever unemployed	−0.014	0.342	3,603	3,603
	(0.064)			
Ever self-employed	0.015	0.206	3,579	3,579
	(0.054)			
Ever part-time	0.023	0.475	3,579	3,579
	(0.057)			
Number of jobs	−0.008	5.294	3,603	3,603
	(0.343)			
Early retirement	0.016	0.533	5,927	5,927
	(0.056)			
C. Health behavior				
Ever smoked	−0.138**	0.606	3,601	3,601
	(0.064)			
Smokes currently	−0.047	0.098	31,285	6,678
	(0.031)			
Drank alcohol last 12 months	−0.038	0.845	27,205	6,487
	(0.038)			
Risky drinking (15+/week)	−0.013	0.056	31,527	6,730
	(0.018)			
Any light activity	0.027	0.874	31,527	6,730
	(0.026)			
Any moderate activity	0.126***	0.755	31,527	6,730
	(0.036)			
Any vigorous activity	0.054	0.304	31,527	6,730
	(0.036)			
BMI	−0.538	27.976	11,426	4,951
	(0.616)			

Note: Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

between the lowest and the fourth quintile of socioeconomic status. Second, I estimate the causal effect of education on old-age frailty by EA PGI in a fuzzy regression discontinuity design, using the 1947 raising of the UK school leaving age from 14 to 15 as exogenous variation in schooling. The health returns to schooling rise monotonically along the EA PGI, from near zero in the lowest quintile to a large 6 percentage point reduction in frailty in the highest. Estimating marginal treatment effects to

account for differences in unobserved complier characteristics that may drive these effects, shows that educational returns in the lowest part of the EA PGI may have been understated and the interaction effect may be lower than suggested by the 2SLS estimation. Nevertheless, by far the largest returns remain concentrated among those in the highest EA PGI quintile.

Overall, I show that an early-life interaction between an exogenous increase in schooling and genetic propensity contributes to larger health differences late in life. Such a complementarity between genes and educational attainment matches the notion that those with a higher genetic predisposition for education extract larger benefits from their education (Houmark, Ronda, and Rosholm 2024). It is also in line with findings that the effect of education on income and on cognitive ability in old age increase in the EA PGI (Barcellos, Carvalho, and Turley 2021; Hollenbach, Schmitz, and Westphal 2026). On the other hand, increasing schooling seems to reduce the impact of genetic *risk* for specific conditions, including overweight and Alzheimer's disease (Barcellos, Carvalho, and Turley 2018; Barcellos et al. 2025). These findings do not conflict. The first refers to skill formation, where only those at the upper end of genetic propensity for schooling can translate additional instruction time later-life outcomes. The second refers to compensating effects of additional education, which could, for example, materialize through better health knowledge.

This study shows that a large-scale schooling policy increased the genetic health gradient by reduced frailty mainly for the top of the EA PGI distribution. Nevertheless, it reduced genetic differences in educational attainment by moving more individuals from the lower part of the EA PGI distribution into additional schooling (see the first-stage results in Appendix Figure A.1 and Hollenbach, Schmitz, and Westphal 2026, who cover this in more detail). Yet, this larger increase in additional schooling did not lead to imrovements in later-life health. This does not imply that additional schooling at other margins, such as tertiary education, could not benefit low-PGI students disproportionately.

I do not interpret my results as evidence that the EA PGI can help target educational policies toward individuals at the lower end or a group that would be able to most effectively utilize the change. Apart from ethical concerns about collection of genetic data from every child, most PGIs are currently constructed from samples of European ancestry and their validity in broader populations is unclear (Martin et al. 2019). Moreover, their predictive power at the individual level is limited (Okbay et al. 2022). Nevertheless, they are a useful, albeit imperfect, proxy for the ability to learn and can show where policies fail or where they are particularly effective in changing behavior and outcomes.

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Appendix A. Additional figures and tables

TABLE A.1 – Frailty index items and coding

Description	Coding
Difficulty with walking 100 yards	No=0, Yes=1
Difficulty sitting for about two hours	No=0, Yes=1
Difficulty getting up from a chair after sitting for long periods	No=0, Yes=1
Difficulty climbing several flights of stairs without resting	No=0, Yes=1
Difficulty climbing one flight of stairs without resting	No=0, Yes=1
Difficulty stooping, kneeling, or crouching	No=0, Yes=1
Difficulty reaching or extending arms above shoulder level	No=0, Yes=1
Difficulty pulling or pushing large objects like a living room chair	No=0, Yes=1
Difficulty lifting or carrying weights over 10 pounds, like a heavy bag	No=0, Yes=1
Difficulty picking up a 5p coin from a table	No=0, Yes=1
Difficulty dressing, including putting on shoes and socks	No=0, Yes=1
Difficulty walking across a room	No=0, Yes=1
Difficulty bathing or showering	No=0, Yes=1
Difficulty eating, such as cutting up your food	No=0, Yes=1
Difficulty getting in or out of bed	No=0, Yes=1
Difficulty using the toilet, including getting up or down	No=0, Yes=1
Difficulty using a map to figure out how to get around in a strange place	No=0, Yes=1
Difficulty preparing a hot meal	No=0, Yes=1
Difficulty shopping for groceries	No=0, Yes=1
Difficulty making telephone calls	No=0, Yes=1
Difficulty taking medications	No=0, Yes=1
Difficulty managing money (e.g. paying bills, keeping track of expenses)	No=0, Yes=1
Difficulty doing work around the house or garden	No=0, Yes=1
Self-reported general health	Excellent=0, v.good=0.25, Good=0.5, Fair=0.75, Poor=1
Felt depressed much of the time during the past week	No=0, Yes=1
Felt everything done during the past week was an effort	No=0, Yes=1
Felt sleep was restless much of the time during the past week	No=0, Yes=1
Was happy much of the time during the past week	Yes=0, No=1
Felt lonely much of the time during the past week	No=0, Yes=1
Enjoyed life much of the time during the past week	Yes=0, No=1
Felt sad much of the time during the past week	No=0, Yes=1
Could not get going much of the time during the past week	No=0, Yes=1
High blood pressure or hypertension	No=0, Yes=1
Angina	No=0, Yes=1
Heart attack (including MI or coronary thrombosis)	No=0, Yes=1
Congestive heart failure	No=0, Yes=1
Abnormal heart rhythm	No=0, Yes=1
Diabetes or high blood sugar	No=0, Yes=1
Stroke (cerebral vascular disease)	No=0, Yes=1
Chronic lung disease (e.g. chronic bronchitis or emphysema)	No=0, Yes=1
Asthma	No=0, Yes=1
Arthritis (including osteoarthritis or rheumatism)	No=0, Yes=1
Osteoporosis	No=0, Yes=1
Cancer or malignant tumor (excluding minor skin cancers)	No=0, Yes=1

Continued on next page

Table A.1 continued from previous page

Description	Coding
Parkinson's disease	No=0, Yes=1
Any emotional, nervous or psychiatric problems	No=0, Yes=1
Alzheimer's disease	No=0, Yes=1
Dementia, organic brain syndrome, senility or other serious memory impairment	No=0, Yes=1
Self-reported eyesight function (with lenses, if appropriate)	Excellent=0, v.good=0.2, Good=0.4, Fair=0.6, Poor=0.8, Blind=1
Self-reported hearing function (with hearing aid, if appropriate)	Excellent=0, v.good=0.25, Good=0.5, Fair=0.75, Poor=1
Identify today's date: day of month	No=0, Yes=1
Identify today's date: month	No=0, Yes=1
Identify today's date: year	No=0, Yes=1
Identify the day of the week	No=0, Yes=1
Immediate word recall (quartiles)	Q1=1, Q2=0.67, Q3=0.33, Q4=0
Delayed word recall (quintiles)	Q1=1, Q2=0.75, Q3=0.5, Q4=0.25, Q5=0

TABLE A.2 – Summary statistics (full table)

	Full sample				By E_i (Left school at 15 or later)		
	Mean	SD	Min	Max	0	1	Diff. (p-val.)
Frailty index	0.162	0.110	0.000	0.772	0.194	0.153	−0.041 (0.000)***
Left school at 15 or later	0.790		0	1	0.000	1.000	1.000
Born 1933 or later	0.685		0	1	0.143	0.829	0.686 (0.000)***
Educational attainment PGI							
1st quintile	0.186				0.234	0.174	−0.061 (0.000)***
2nd quintile	0.199				0.214	0.195	−0.019 (0.008)***
3rd quintile	0.200				0.189	0.203	0.014 (0.048)**
4th quintile	0.209				0.206	0.210	0.004 (0.582)
5th quintile	0.206				0.158	0.219	0.061 (0.000)***
Female	0.544		0	1	0.525	0.549	0.024 (0.009)***
Age	74.238	5.344	65.000	85.000	77.049	73.491	−3.559 (0.000)***
Birth year	1,935.027	5.241	1,923.000	1,943.000	1,929.798	1,936.416	6.619 (0.000)***
Age mother left education							
≤ 14	0.678				0.656	0.683	0.027 (0.001)***
15	0.060				0.018	0.071	0.053 (0.000)***
16	0.039				0.007	0.048	0.040 (0.000)***
17	0.013				0.002	0.016	0.015 (0.000)***
18	0.017				0.004	0.020	0.016 (0.000)***
≥ 19	0.019				0.005	0.022	0.017 (0.000)***
Missing	0.175				0.308	0.140	−0.168 (0.000)***
Age father left education							
≤ 14	0.663				0.634	0.671	0.037 (0.000)***
15	0.054				0.022	0.062	0.039 (0.000)***
16	0.044				0.005	0.054	0.049 (0.000)***
17	0.010				0.004	0.011	0.008 (0.000)***
18	0.019				0.005	0.022	0.018 (0.000)***
≥ 19	0.030				0.005	0.037	0.031 (0.000)***
Missing	0.181				0.325	0.143	−0.182 (0.000)***
Survey wave							
1	0.104				0.188	0.081	−0.107 (0.000)***
2	0.125				0.189	0.108	−0.082 (0.000)***
3	0.124				0.164	0.114	−0.051 (0.000)***
4	0.146				0.152	0.145	−0.008 (0.226)
5	0.131				0.118	0.135	0.017 (0.006)***
6	0.114				0.087	0.122	0.035 (0.000)***
7	0.093				0.055	0.103	0.048 (0.000)***
8	0.075				0.030	0.087	0.057 (0.000)***
9	0.058				0.012	0.071	0.059 (0.000)***
10	0.029				0.004	0.036	0.032 (0.000)***
PC1	−0.000	0.012	−0.027	0.041	−0.001	−0.000	0.000 (0.120)
PC2	0.000	0.012	−0.027	0.030	−0.000	0.000	0.000 (0.039)**
PC3	0.000	0.011	−0.057	0.047	0.000	0.000	0.000 (0.261)
PC4	0.000	0.012	−0.036	0.054	0.000	0.000	−0.000 (0.543)
PC5	0.000	0.012	−0.039	0.040	0.000	0.000	−0.000 (0.088)*
PC6	0.000	0.012	−0.034	0.044	−0.001	0.000	0.001 (0.000)***
PC7	0.000	0.012	−0.034	0.042	−0.001	0.000	0.001 (0.000)***
PC8	0.000	0.012	−0.039	0.042	0.000	−0.000	−0.000 (0.186)
PC9	0.000	0.012	−0.039	0.040	−0.000	0.000	0.000 (0.068)*
PC10	−0.000	0.012	−0.048	0.048	−0.000	−0.000	0.000 (0.603)
Observations			18,705		3,927	14,778	

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

TABLE A.3 – Reduced-form and first-stage estimates

	DV: Left school at 15 or later (E_i)	F
$Z_i \times (G_i = 1)$	0.653***	221.9
	(0.044)	
$Z_i \times (G_i = 2)$	0.548***	159.2
	(0.043)	
$Z_i \times (G_i = 3)$	0.487***	122.9
	(0.044)	
$Z_i \times (G_i = 4)$	0.447***	103.3
	(0.044)	
$Z_i \times (G_i = 5)$	0.355***	64.9
	(0.044)	
Observations	18,705	
Individuals	3,415	

Note: First stages estimated according to Eq. 3 on the main sample described in Section 3.1. Each interaction of the instrument Z_i with a PGI quintile G_i yields a separate first stage (though estimated in one model). The second column reports F-statistics for the excluded instrument, separately for each interaction. Controls: linear cohort trend, its interaction with the instrument, indicator variables for gender, survey wave, age, and mother's and father's age when they left full-time education, and the first ten principal components of the genetic data. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

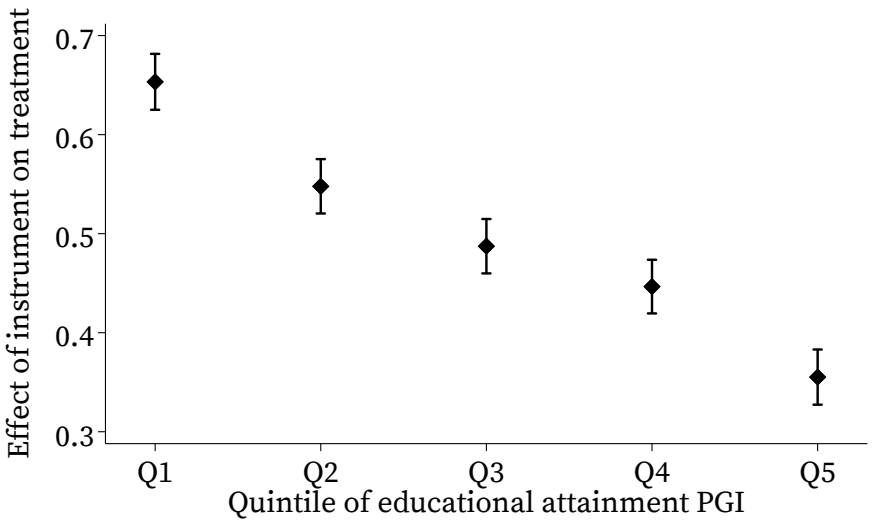


FIGURE A.1 – First-stage estimates by EA PGI quintile

Note: Coefficients π_g of Equation 3 with 95% confidence intervals, which correspond to complier shares by EA PGI quintile. The instrument is being born in or after 1933, the treatment is staying in school until at least age 15. Controls: linear cohort trend, its interaction with the instrument, indicator variables for gender, survey wave, age, and mother's and father's age when they left full-time education, and the first ten principal components of the genetic data. Appendix Table A.3 reports the exact estimates, standard errors, and first-stage F-statistics. Standard errors clustered at the individual level. Estimation sample as described in Section 3.2 with 18,705 observations from 3,415 individuals.

TABLE A.4 – Panel attrition

DV: Attrition	(1) 2SLS	(2) RF
E_i	−0.007	
	(0.037)	
Z_i		−0.003
		(0.020)
Controls	Yes	Yes
Observations	25,805	25,805
Individuals	3,415	3,415

Note: Notes: The dependent variable is an indicator equal to one if an individual eligible for interview in a given wave does not respond. E_i is the education indicator (stayed in school until at least age 15), instrumented by Z_i in column (1). Z_i is an indicator for exposure to the 1947 compulsory schooling reform (born in 1933 or later). Column (1) reports the 2SLS estimate, column (2) the reduced-form estimate. All specifications control for the running variable, its interaction with Z_i, gender, and wave fixed effects. Standard errors clustered at the individual level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

Appendix B. Marginal treatment effects: framework and technical details

This appendix complements the estimation of marginal treatment effects (MTEs) in Section 4.5. Section B.1 introduces the MTE framework more formally. Section B.2 describes how I estimate the type shares and type-specific average outcomes that the MTE curves must reproduce. Section B.3 uses them to estimate linear MTRs and to test for essential heterogeneity, which motivates the shape restrictions. Section B.4 briefly introduces Bernstein polynomials. Appendix Section C then describes the estimation of the bounds step by step. For a comprehensive treatment, see Mogstad, Santos, and Torgovitsky (2018) and Mogstad and Torgovitsky (2024), and the Online Appendix F of Hollenbach, Schmitz, and Westphal (2026) for an application to gene-environment interactions in the same setting.

B.1 A brief introduction to marginal treatment effects

Treatment choice and unobserved resistance. I follow the formulation in Online Appendix F.1 of Hollenbach, Schmitz, and Westphal (2026), which builds on Heckman and Vytlacil (2005). Potential frailty with and without staying in school until at least age 15 (the treatment) depends on the EA PGI, observable characteristics, and an unobserved component,

$$Y_i^j(G_i) = \mu_j(G_i, X_i) + \varepsilon_i^j(G_i), \qquad j \in \{0, 1\}.$$

In a generalized Roy model (Roy 1951), individuals stay in school if the expected returns exceed the monetary and non-monetary costs $C_i = \mu_C(G_i, X_i, Z_i) + U_i^C$. The costs depend on the EA PGI, the controls, the instrument, and an unobserved term. The instrument affects the decision only through the costs and does not enter the potential outcomes. The decision rule is

$$\begin{aligned} E_i = 1 \;&\Leftrightarrow\; Y_i^1(G_i) - Y_i^0(G_i) > C_i \\ &\Leftrightarrow\; \underbrace{\mu_1(G_i, X_i) - \mu_0(G_i, X_i) - \mu_C(G_i, X_i, Z_i)}_{\mu_E(G_i, X_i, Z_i)} \;>\; \underbrace{-\left(\varepsilon_i^1(G_i) - \varepsilon_i^0(G_i) - U_i^C\right)}_{V_i(G_i)}. \end{aligned}$$

The EA PGI therefore enters the choice in two ways. It shifts the observable part of the expected net returns, μ_E, for example because students with a higher EA PGI acquire skills more effectively or face lower costs of schooling, and it can interact with the instrument, so that the reform shifts the decision differently in different quintiles. This is what produces the first-stage gradient in Appendix Figure A.1. The unobserved term $V_i(G_i)$ collects everything else, such as preferences for schooling or perceived costs. Because it contains the unobserved parts of the returns, E_i is correlated with the unobserved part of the potential outcomes, which makes it endogenous. Applying the cumulative distribution function of $V_i(G_i)$ to both sides of the inequality gives the equivalent decision rule

$$E_i = \mathbb{1}\left[\, p(G_i, X_i, Z_i) \geq U_i^D \right],$$

where $p(G_i, X_i, Z_i) = \Pr(E_i = 1 \mid G_i, X_i, Z_i)$ is the propensity score and U_i^D is uniformly distributed on the unit interval within each quintile. U_i^D is the percentile rank of individual i's unobserved resistance

to schooling. An individual with $U_i^D = 0.1$ is more willing to stay in school than 90 percent of the individuals in their quintile. An individual stays in school whenever the propensity score exceeds their resistance.

Response types. With a binary instrument and controls held fixed, the propensity score in quintile g takes two values: $p_g(0)$, the share that stays in school without the reform, and $p_g(1)$, the share that stays in school with it. This divides the unit interval into the three response types:

$$\text{always-takers: } U_i^D \in [0, p_g(0)], \qquad \text{compliers: } U_i^D \in (p_g(0), p_g(1)], \qquad \text{never-takers: } U_i^D \in (p_g(1), 1].$$

This ordering is implied by the monotonicity assumption (no defiers) of Section 3.4. The complier share $p_g(1) - p_g(0)$ is the first stage π_g. Figure 5 shows the resulting intervals by quintile.

MTEs, MTRs, and target parameters. The MTE is the average effect of the treatment for individuals at a given resistance,

$$\text{MTE}_g(u) = \mathbb{E}\big[Y_i^1 - Y_i^0 \mid U_i^D = u, G_i = g\big] = m_g^1(u) - m_g^0(u),$$

and consists of the two marginal treatment response (MTR) curves, the average potential outcomes at that resistance,

$$m_g^j(u) = \mathbb{E}\big[Y_i^j \mid U_i^D = u, G_i = g\big], \qquad j \in \{0,1\},\ g \in \{1,\dots,5\}.$$

Standard treatment parameters are averages of the MTE over different intervals of U^D (Heckman and Vytlacil 2005). The average treatment effect in quintile g averages over the entire unit interval, the LATE estimated by 2SLS only over the complier interval:

$$\text{ATE}_g = \int_0^1 \text{MTE}_g(u)\,du, \qquad \beta_g = \text{LATE}_g = \frac{1}{p_g(1) - p_g(0)} \int_{p_g(0)}^{p_g(1)} \text{MTE}_g(u)\,du.$$

The 2SLS interaction effect $\beta_5 - \beta_1$ therefore compares averages over two different intervals. It coincides with the difference in effects at a common resistance only if the MTEs are flat (no essential heterogeneity) or the complier intervals coincide. I bound three types of target parameters, each in its own optimization (Appendix Section C):

i. **The MTE curves**, $\text{MTE}_g(u)$, separately at each of 21 equally spaced points u and for each quintile (Figure 6).
ii. **The average MTE within each quintile** over the interval $[a, b] = [0.55, 0.85]$, which is populated by compliers from all quintiles (Panel A of Table 4):

$$\bar{\tau}_g = \frac{1}{b-a} \int_a^b \text{MTE}_g(u)\,du.$$

iii. **The interaction effect** between the highest and the lowest quintile over the same interval (Panel B of Table 4):

$$\bar{\tau}_5 - \bar{\tau}_1 = \frac{1}{b-a} \int_a^b \left[\text{MTE}_5(u) - \text{MTE}_1(u) \right] du.$$

For frailty, a negative MTE is a health improvement. A non-increasing MTE therefore means that the health gains from schooling are at least as large for individuals with high resistance as for those with low resistance.

Average outcomes from the data. The data do not reveal the MTR curves themselves, only averages over the type intervals. In each quintile, four such averages are identified (Imbens and Rubin 1997):

$$\mathbb{E}[Y_i^1 \mid AT, g] = \frac{1}{p_g(0)} \int_0^{p_g(0)} m_g^1(u)\, du, \qquad \mathbb{E}[Y_i^1 \mid C, g] = \frac{1}{p_g(1) - p_g(0)} \int_{p_g(0)}^{p_g(1)} m_g^1(u)\, du,$$

$$\mathbb{E}[Y_i^0 \mid C, g] = \frac{1}{p_g(1) - p_g(0)} \int_{p_g(0)}^{p_g(1)} m_g^0(u)\, du, \quad \mathbb{E}[Y_i^0 \mid NT, g] = \frac{1}{1 - p_g(1)} \int_{p_g(1)}^{1} m_g^0(u)\, du.$$

Infinitely many curves reproduce these four average outcomes. Moreover, the data contain no information on the untreated outcome of always-takers or the treated outcome of never-takers, so the MTE in these regions is determined by extrapolation, and therefore functional form of the curves. Furthermore, without a continuous instrument that would trace out the curves over the entire complier range, the MTEs cannot be not point identified. Mogstad, Santos, and Torgovitsky (2018) show how to bound any treatment parameter that is a weighted average of MTRs by searching over all curves that are consistent with the average outcomes and with additional restrictions. With a binary instrument, the main alternative is to assume linear MTRs, which are then exactly identified by the two averages per treatment state (Brinch, Mogstad, and Wiswall 2017). I use this linear version as a benchmark and test for essential heterogeneity in Section B.3, but avoid it for estimating treatment effects, since linearity is a strong assumption.

B.2 Type shares and average outcomes by quintile

All quantities in this section are identified under the standard LATE assumptions and follow Imbens and Rubin (1997). Appendix G of Hollenbach, Schmitz, and Westphal (2026) provides a detailed step-by-step derivation. Two regressions provide everything needed. For the type shares, I estimate a linear probability model of the treatment on a full set of interactions between the instrument and the PGI quintile indicators, without an intercept:

$$E_i = \sum_{g=1}^{5} \left(\zeta_{0g}\, \mathbb{1}[Z_i = 0]\, \mathbb{1}[G_i = g] + \zeta_{1g}\, \mathbb{1}[Z_i = 1]\, \mathbb{1}[G_i = g] \right) + X_i' \eta + \kappa_i. \tag{4}$$

For the average outcomes, I regress the frailty index on a full set of interactions between treatment, instrument, and quintile, again without an intercept:

$$Y_{it} = \sum_{g=1}^{5} \sum_{e,z \in \{0,1\}} \lambda_{ezg} \, \mathbb{1}[E_i = e] \, \mathbb{1}[Z_i = z] \, \mathbb{1}[G_i = g] + X_{it}' \mu + \nu_{it}. \tag{5}$$

X_{it} contains the same controls as Equation 2: the running variable and its interaction with the instrument, and indicator variables for gender, survey wave, age, and mother's and father's school-leaving age, as well as the first ten genetic principal components. The controls enter additively with common coefficients, and I read the cell means directly off the coefficients, that is, at the cutoff ($c_i = 0$) and at the reference values of the remaining controls. Because the controls enter additively, they shift all cell means of a quintile by the same amount. They therefore affect the level of the MTR curves but not the differences between treated and untreated outcomes that make up the treatment effects.

The coefficients map into the type shares and average outcomes as follows:

i. **Shares of always-takers and never-takers.** Individuals who take the treatment although the instrument is switched off must, under monotonicity, be always-takers, so $\phi_{AT,g} = p_g(0) = \zeta_{0g}$. Analogously, those who do not take the treatment although the instrument is switched on must be never-takers, so $\phi_{NT,g} = 1 - p_g(1) = 1 - \zeta_{1g}$.

ii. **Share of compliers.** Since the three shares add up to one, $\phi_{C,g} = p_g(1) - p_g(0) = \zeta_{1g} - \zeta_{0g}$, the first stage π_g.

iii. **Treated outcome of always-takers and untreated outcome of never-takers.** The cells $E_i = 1, Z_i = 0$ and $E_i = 0, Z_i = 1$ consist only of always-takers and never-takers, respectively. Under instrument exogeneity, their means identify $\mathbb{E}[Y_i^1 \mid AT, g] = \lambda_{10g}$ and $\mathbb{E}[Y_i^0 \mid NT, g] = \lambda_{01g}$.

iv. **Treated and untreated outcome of compliers.** The cell $E_i = 1, Z_i = 1$ mixes always-takers and compliers, and the cell $E_i = 0, Z_i = 0$ mixes never-takers and compliers. Solving for the complier part gives

$$\mathbb{E}[Y_i^1 \mid C, g] = \lambda_{11g} \frac{\phi_{AT,g} + \phi_{C,g}}{\phi_{C,g}} - \lambda_{10g} \frac{\phi_{AT,g}}{\phi_{C,g}}, \tag{6}$$

$$\mathbb{E}[Y_i^0 \mid C, g] = \lambda_{00g} \frac{\phi_{NT,g} + \phi_{C,g}}{\phi_{C,g}} - \lambda_{01g} \frac{\phi_{NT,g}}{\phi_{C,g}}. \tag{7}$$

Their difference is the LATE in quintile g.

For the pooled test for essential heterogeneity in Section B.3, I estimate the same two regressions without interacting with the PGI quintiles and evaluate the controls at their sample means.

B.3 Linear MTRs and a test for essential heterogeneity

The three types can be ordered along U^D: always-takers between 0 and ϕ_{AT}, compliers between ϕ_{AT} and $\phi_{AT} + \phi_C$, and never-takers between $\phi_{AT} + \phi_C$ and 1. Following Brinch, Mogstad, and Wiswall (2017), I treat each type-specific mean as if it were measured in the middle of that type's interval. With a binary instrument, there are only two means per potential outcome: always-takers and compliers

when treated, compliers and never-takers when untreated. Assuming that the MTRs are linear in U^D, the line through the two means identifies the MTR. Its slope is the difference in means divided by the horizontal distance between the two midpoints, which is half the sum of the two type shares. A plain difference in means would ignore how far apart the groups are on the unit interval. The same difference in frailty is a much steeper change between two narrow groups than between two wide ones. I calculate:

i. **Slope of the treated MTR**, from the two treated means:

$$\frac{\mathbb{E}[Y_i^1 \mid C] - \mathbb{E}[Y_i^1 \mid AT]}{(\phi_{AT} + \phi_C)/2}.$$

ii. **Slope of the untreated MTR**, from the two untreated means:

$$\frac{\mathbb{E}[Y_i^0 \mid NT] - \mathbb{E}[Y_i^0 \mid C]}{(\phi_C + \phi_{NT})/2}.$$

iii. **Slope of the linear MTE**, the difference of the two slopes above. The MTE at any u is the gap between the treated and the untreated MTR, so the difference in slopes shows whether this gap widens or narrows as resistance increases. A slope of zero means that treatment effects do not depend on resistance to treatment, that is, there is no essential heterogeneity. Since frailty is a bad outcome, a positive slope would imply selection on gains: those most willing to stay in school would experience the largest reductions in frailty. A negative slope implies the reverse, with the largest reductions among those with the highest resistance.

Table B.1 reports the type shares, the four means, the differences in means, the slopes of the linear MTRs, and the test for essential heterogeneity. All differences and slopes are small and not statistically significant. Both linear MTRs have positive slopes: with and without the additional year, frailty tends to increase with resistance to schooling. The untreated MTR increases more steeply, mainly because never-takers are somewhat frailer than untreated compliers. The implied MTE slope of −0.028 is therefore negative but not statistically significant. These results guide the shape restrictions of the bounding approach in Section 4.5. I restrict both MTRs to be non-decreasing and the MTE to be non-increasing in U^D. Since all three restrictions are weak, they include the case of flat curves and thus of no essential heterogeneity, which the test cannot reject.

Standard errors. I obtain standard errors by resampling rather than analytically. I use the Bayesian bootstrap of Rubin (1981) with 1,000 replications. In each replication, every individual receives a weight drawn from an exponential distribution with mean one, which is constant across all waves of that individual. The full procedure, both regressions, the type shares, the complier means, and the slopes, is re-estimated with each set of weights, and the reported standard error is the standard deviation of the resulting estimates. Unlike the classic bootstrap, the Bayesian bootstrap never drops individuals from a replication, so no cell runs empty, and the estimates are smooth in the weights, which is convenient because the quantities of interest involve ratios of estimated shares.

TABLE B.1 – Frailty by response type: potential outcomes and test for essential heterogeneity

Type	Share	Potential outcome	Mean frailty	Difference of means	Slope in potential outcomes	Selection on gains
Always-taker	0.431	Treated AT	0.157	0.004	0.010	
		Treated C	0.162	(0.015)	(0.031)	
Complier	0.495					-0.028
		Untreated C	0.189	0.011	0.038	(0.051)
Never-taker	0.074	Untreated NT	0.200	(0.015)	(0.053)	

Note: Shares of always-takers (AT), compliers (C), and never-takers (NT) alongside treated potential outcome means for always-takers and compliers, untreated potential outcome means for compliers and never-takers, differences in means, slopes of linear MTRs, and the test for essential heterogeneity (selection on gains), which is the difference between the two slopes. All estimates use the main sample described in Section 3.1 and are pooled across EA PGI quintiles. The shares come from a first-stage regression of the treatment (in school until at least 15) on the instrument (born 1933 or later). The means are based on a regression of the frailty index on interactions of the treatment with the instrument. Appendix Sections B.2 and B.3 describe how each number is calculated. Standard errors are computed using the Bayesian bootstrap (Rubin 1981) with 1,000 replications and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ indicate significance levels.

B.4 Bernstein polynomials

I approximate each MTR by a Bernstein polynomial, as in Mogstad, Santos, and Torgovitsky (2018) and Hollenbach, Schmitz, and Westphal (2026). A Bernstein polynomial of degree K on the unit interval is a weighted sum of $K + 1$ basis functions:

$$m_g^j(u) = \sum_{v=0}^{K} \theta_v^{jg}\, b_{v,K}(u), \qquad b_{v,K}(u) = \binom{K}{v} u^v (1-u)^{K-v}.$$

Figure B.1 shows the seven basis functions for $K = 6$, the degree I use. Each basis function peaks at $u = v/K$, so the coefficient θ_v^{jg} mainly decides the height of the curve around that point. Bernstein polynomials are flexible enough to approximate almost any smooth shape on the unit interval. Their main advantage, however, is that all relevant restrictions become linear conditions on the coefficients. Since the basis functions are non-negative and sum to one at every u, the curve at any point is a weighted average of its coefficients. If all coefficients lie between 0 and 1, so does the curve. If the coefficients are non-decreasing in v, the curve is non-decreasing in u. Averages of the curve over an interval, such as the type-specific means, are also linear in the coefficients. Finding the smallest and largest value of a target parameter over all admissible curves can therefore be framed as a linear programming problem (Mogstad, Santos, and Torgovitsky 2018). With $K = 6$, each MTR has seven coefficients, so the ten MTRs have 70 coefficients in total. A higher degree allows more flexible shapes but makes the bounds wider, because more curves are consistent with the same four averages per quintile.

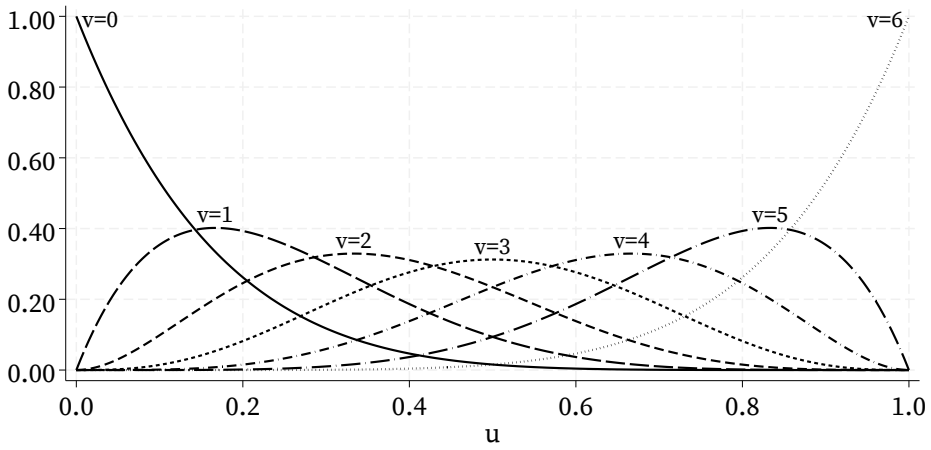


FIGURE B.1 – Bernstein basis functions of degree six

Note: The seven basis functions $b_{\nu,6}(u) = \binom{6}{\nu} u^{\nu}(1-u)^{6-\nu}$, $\nu = 0, \ldots, 6$, that compose a Bernstein polynomial of degree six on the unit interval. Each MTR is a weighted sum of these functions, with the weights θ_{ν} chosen by the linear program.

Appendix C. Estimating the MTE bounds step by step

This section describes how I obtain the bounds in Figure 6 and Table 4. For exposition, I present the procedure as a sequence of steps. In practice, steps 2 to 5 are solved together: each bound is a single linear program over all ten MTRs, with the data and the restrictions as constraints. The procedure follows Mogstad, Santos, and Torgovitsky (2018) and is similar to Online Appendix F of Hollenbach, Schmitz, and Westphal (2026), which provides a more detailed computation protocol. It differs in three respects: the shape restrictions (step 3), the fit of the curves to the data up to a tolerance rather than exactly (step 4), and separate optimizations for the effects within each quintile and for the interaction effect (step 5).

Step 1: Type shares and average outcomes. In each EA PGI quintile, I estimate the propensities $p_g(0)$ and $p_g(1)$ and the four type-specific average outcomes, $\mathbb{E}[Y_i^1 \mid AT, g]$, $\mathbb{E}[Y_i^1 \mid C, g]$, $\mathbb{E}[Y_i^0 \mid C, g]$, and $\mathbb{E}[Y_i^0 \mid NT, g]$, as described in Appendix Section B.2. This yields ten propensities, which fix the type intervals on the unit interval, and 20 average outcomes, which I refer to as moments.

Step 2: Parameterization. I approximate each of the ten MTRs $m_g^j(u)$, two treatment states times five quintiles, by a Bernstein polynomial of degree six (Appendix Section B.4). The 70 coefficients θ_v^{jg} are the unknowns of the linear program. Every quantity below, an MTR at a point, its average over an interval, or the MTE, is a linear function of these coefficients.

Step 3: Shape restrictions. I restrict the shape of the curves in four ways. Each restriction is imposed on the polynomial coefficients, which is straightforward because the shape of a Bernstein polynomial follows the shape of its coefficients (Appendix Section B.4).

(a) **Range.** All MTRs lie between 0 and 1, the range of the frailty index.

(b) **Non-decreasing MTRs.** With and without the additional year of schooling, frailty does not decrease with resistance to schooling. Individuals who are more inclined to stay in school are, on average, at most as frail as those who are less inclined. The restriction is weak, so flat curves are allowed. It is motivated by the positive slopes of both linear MTRs in Appendix Table B.1 and corresponds to the assumption of weakly monotone untreated outcomes in Kowalski (2023), which I extend to treated outcomes.

(c) **Non-increasing MTE.** The effect of schooling on frailty does not increase with resistance to schooling. Again, flat MTEs, that is, no essential heterogeneity, are allowed. The restriction is motivated by the negative, though not statistically significant, slope of the linear MTE in Appendix Table B.1. For frailty, it rules out selection on gains.

(d) **Similar shapes across quintiles.** The MTRs may differ in level across quintiles, but their shapes must be similar to those in the lowest quintile. Hollenbach, Schmitz, and Westphal (2026) impose identical shapes, which corresponds to the additive separability commonly assumed in MTE estimation. I allow the shapes to differ moderately, so that the interaction effect may vary with U^D, while information on the shape of the curves is still shared across quintiles.

The monotonicity restrictions in (b) and (c) hold up to a small tolerance, which avoids numerical problems.

Step 4: Fit to the data. The MTRs must reproduce the 20 moments from step 1 when averaged over the respective type intervals. Since the moments are estimated with sampling noise, requiring an exact match can be infeasible or let the noise drive the bounds. Following Mogstad, Santos, and Torgovitsky (2018, Section 3.4), I therefore proceed in two stages. First, I measure the misfit of a set of curves as the sum over all quintiles and types of the absolute difference between the curve's average over the type interval and the estimated moment, weighted by the width of the interval. The misfit is thus an average deviation over the unit interval. I compute the smallest misfit Q^* that any set of curves satisfying the restrictions in step 3 can achieve. Second, I retain all curves whose misfit is at most 5 percent above Q^*. The restrictions in step 3 always hold exactly (up to the stated tolerances); only the fit to the noisy moments is relaxed.

Step 5: Bounds. Among all curves that satisfy steps 3 and 4, I search for those that minimize and those that maximize each of the target parameters listed in Appendix Section B.1. Each bound is a separate linear program. For the average MTE over $[0.55, 0.85]$, this requires one minimization and one maximization per quintile. For the interaction effect, I directly minimize and maximize $\bar{\tau}_5 - \bar{\tau}_1$. This is not the same as subtracting the quintile-specific bounds, which would combine the curves that minimize the effect in one quintile with those that maximize it in another, even though these need not be jointly admissible. Subtracting bounds would therefore yield an interval that is too wide. All quintiles enter every linear program, since the restrictions in (d) and the misfit in step 4 link the curves across quintiles. For Figure 6, I repeat the minimization and maximization of $\text{MTE}_g(u)$ separately at each of 21 equally spaced points of u, which yields pointwise bounds on the MTE curves. The bounds on the average MTE in Table 4 are not averages of these pointwise bounds, since each point may be attained by different curves.

Step 6: Inference. I obtain standard errors from a Bayesian bootstrap (Rubin 1981) with 300 replications. In each replication, every individual receives a weight drawn from an exponential distribution with mean one, constant across all waves of that individual. With these weights, I re-estimate the moments in step 1 and re-solve steps 4 and 5, including the best fit Q^*. The standard error of each bound is its standard deviation across replications. Stars refer to a one-sided test that a bound is zero against the alternative that it has the sign it has, $p = \Phi(-|\text{bound}/\text{SE}|)$. When both bounds are negative, a significant minimum, the bound closest to zero, implies that the entire identified set lies significantly below zero. Since bounds are minima and maxima, they are not smooth functions of the data, so the standard errors should be regarded as approximate. I additionally compute 95 percent confidence intervals for the effect itself following Imbens and Manski (2004), which account for both bounds. They exclude zero in quintiles 2 and 5 and include it for the interaction effect.